Applied Soft Computing Journal 202 (2026) 115742

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

BERTidRAAF for AI-driven recruitment: Rare-aware adaptive fusion of BERT-based context and IDF-weighted keywords for CV classification

Omar Haddad, Mohamed Nazih Omri*

MARS Research Lab LR17ES05, University of Sousse, Sousse, Tunisia

HIGHLIGHTS

- Introduces BERTidRAAF, a rare-aware BERT-IDF architecture for AI-driven resume classification.
- Injects training-only IDF evidence after BERT contextualization to preserve rare professional terms.
- Uses rare-aware gating, multi-head fusion, and validation-selected decision policy for auditable evaluation.
- Validates Resume and FairCVdb results with ablations, fairness checks, efficiency analysis, and rare-term diagnostics.

ARTICLE INFO

Keywords:
Resume classification
Natural language processing
BERT
IDF weighting
Rare-aware fusion
Recruitment automation
FairCVdb
Reproducible benchmarking

ABSTRACT

Resume classification is a demanding long-document problem because the target label is often determined by a small number of highly informative professional expressions, certifications, tools, and domain-specific job titles embedded in heterogeneous career narratives. Contextual transformers capture semantic regularities, whereas inverse document frequency (IDF) preserves the salience of rare terms; however, the two sources of evidence are often combined either before contextualization or through weak late fusion. This paper presents BERTidRAAF, a rare-aware adaptive fusion framework that injects training-corpus IDF evidence after BERT contextualization. The model aligns IDF scores with WordPiece tokens, computes contextual and IDF-weighted summaries, learns a gate between global and rare-term representations, combines complementary prediction heads, and selects the final decision policy on a validation split before held-out testing. The evaluation uses a 24-class Resume dataset with 1785/199/497 train/validation/test instances and an external binary FairCVdb setting with 17,280/1920/4800 instances. On Resume, BERTidRAAF reaches 85.71% accuracy, 0.8038 macro-F1, and 0.9802 macro-AUC, improving over the strongest baseline, BERTLARGE, by 3.22 percentage points in accuracy and 0.0443 macro-F1. On FairCVdb, it obtains 96.17% accuracy and 0.9617 macro-F1, slightly higher than BERTLARGE under the same protocol. The study also reports confidence intervals, per-class behavior, error pairs, close BERT-IDF ablations, validation-selected fusion details, and efficiency indicators. The results support the usefulness of post-contextual rare-term weighting for fine-grained resume classification while showing that this accuracy-oriented design increases computational cost. The complete reproducibility material is made available through the companion GitHub repository reported in the Data and Code Availability statement. The extended rare-term recovery diagnostic reports recall@5 = 0.7463 for the IDF branch, compared with 0.7007 for uniform weighting and 0.7183 for attention-based ranking, supporting improved sensitivity to discriminative rare professional terms under the reported salience protocol.

1. Introduction

1.1. Context and motivation

Digital recruitment platforms, public institutions, and human-resource departments increasingly use automated text processing to organize large volumes of resumes and to support candidate-screening workflows [1–3]. Resume classification is more difficult than ordinary short-text classification because resumes are long, semi-structured, and lexically heterogeneous. A single document may contain education, professional experience, sector-specific skills, certifications, tools, project descriptions, and administrative details. The category label is therefore

* Corresponding author.
Email addresses: omarhaddad@fsegs.u-sfax.tn (O. Haddad), mohamednazih.omri@eniso.u-sousse.tn (M.N. Omri).

https://doi.org/10.1016/j.asoc.2026.115742
Received 10 December 2025; Received in revised form 9 June 2026; Accepted 10 June 2026
Available online 15 June 2026

often determined not by the dominant topic of the document, but by a small number of rare yet decisive expressions.

This property creates a central modeling tension. Lexical models such as TF-IDF with linear classifiers preserve explicit keyword evidence and remain efficient [4], but they are limited in their ability to represent contextual equivalence, paraphrase, and long-range semantic relations. Transformer encoders such as BERT capture contextual meaning and have become strong text-classification backbones [5–7]; however, standard pooling can dilute isolated skill names, certifications, tools, and job titles when long resume sequences contain many generic words. A strong resume classifier should therefore preserve contextual semantics while making rare professional evidence visible after contextualization.

1.2. Problem statement

Let $\mathcal{D} = \{(d_i, y_i)\}_{i=1}^{N}$ denote a corpus of resumes and professional-category labels, where each document d_i is a variable-length sequence and $y_i \in \{1, \ldots, C\}$. The objective is to learn a classifier $f(d_i) = \hat{y}_i$ that generalizes to held-out resumes while making the sources of evidence methodologically traceable. The difficulty is that two signals must be combined without leakage: contextual evidence, which captures the semantic role of each token in the resume, and rarity evidence, which highlights low-frequency terms whose discriminative value is high for particular categories.

The research gap addressed in this work is not the isolated use of BERT or IDF. Both are established components. The gap lies in how rarity is injected, controlled, and evaluated. Many hybrid methods concatenate lexical and neural vectors, add convolutional layers on top of transformer embeddings, or use TF-IDF as an external feature space [8,9]. Such strategies do not always isolate post-contextual IDF weighting, evaluate close BERT-IDF ablations, or separate validation-based policy selection from held-out test reporting. These points are especially important in recruitment settings, where weak validation practices can lead to inflated claims and where fairness, accountability, and human oversight remain critical concerns [10–12].

To address fairness concretely, this study adopts the disparate impact ratio (DIR) as the primary fairness evaluation standard, following the AI recruitment auditing framework of the European Commission's ALTAI (Assessment List for Trustworthy AI). A classifier is considered fair if the DIR between any two demographic groups (e.g., gender) is at least 0.8, corresponding to the "four-fifths rule" used in employment jurisprudence. This standard is applied post hoc to the predictions of BERTidRAAF on FairCVdb, where demographic attributes are available.

1.3. Proposed direction

This article presents BERTidRAAF, a BERT-IDF rare-aware adaptive fusion architecture for resume classification. Methodologically, IDF scores are estimated only on the training split, projected to WordPiece positions, and applied after BERT contextualization.

The model then constructs contextual, max-pooled, and IDF-weighted summaries; learns a rare-aware gate; produces multi-head logits; and finalizes the prediction policy on the validation split.

From a validation-protocol perspective, no test data are used for policy selection. From a recruitment-relevance perspective, the architecture is designed to preserve rare professional terms (skills, tools, certifications) that are often decisive in screening decisions.

1.4. Contributions

The article makes the following contributions, separated into three categories.

1.4.1. Methodological architecture

- It introduces BERTidRAAF, a rare-aware BERT-IDF architecture in which corpus-level rarity is injected *after* contextual encoding rather than being treated only as an external bag-of-words feature.
- It provides a six-phase mathematical and algorithmic formulation covering training-only IDF estimation, WordPiece-level IDF projection, masked contextual pooling, IDF-weighted pooling, rare-aware gating, multi-head logit fusion, and validation-selected prediction.
- It aligns the article's formulation with the executed benchmark implementation, including positive softplus head weights, multi-sample dropout, label-smoothed cross-entropy, and validation-controlled policy selection.

1.4.2. Validation protocol contribution

- It introduces a clean separation between IDF estimation (training only), hyperparameter selection (validation only), and final reporting (held-out test), with a validation-selected prediction policy fixed before test evaluation.
- It evaluates the method on a 24-class Resume dataset and an external binary FairCVdb setting using explicit train/validation/test partitions and the same held-out reporting logic.

1.4.3. Recruitment automation relevance

- It compares the proposed model against lexical, neural, transformer, and close BERT-IDF hybrid families under a protocol designed for auditability in recruitment contexts.
- It reports confidence intervals, class-level behavior, confusion pairs, fusion choices, and efficiency indicators to support transparent deployment in screening pipelines.

1.5. Research questions

The empirical study is guided by four research questions:

1. **RQ1:** Does BERTidRAAF improve held-out resume classification performance compared with the strongest evaluated baselines?
2. **RQ2:** Does the proposed model remain competitive on the external FairCVdb binary setting without using test data for policy selection?
3. **RQ3:** Do close BERT-IDF ablations confirm that the complete rare-aware adaptive design is more effective than simpler BERT + IDF, BERT + CNN, or BERT-IDF sequence modules?
4. **RQ4:** What are the class-level, error-pattern, ranking, and computational trade-offs associated with the proposed model?

1.6. Paper organization

The remainder of this paper separates three contribution threads: methodological architecture, validation protocol, and recruitment automation relevance. Section 2 reviews lexical, neural, transformer, and hybrid approaches for resume-oriented text classification. Section 3 presents the proposed BERTidRAAF methodological architecture, including its six-phase pipeline and mathematical formulation. Section 4 describes the validation protocol: datasets, baselines, training, validation splits, metrics, and the explicit separation between IDF estimation (training only), policy selection (validation only), and held-out test reporting. Section 5 analyzes the results, including close BERT-IDF ablations and validation-selected fusion strategies. Section 6 interprets findings for recruitment automation relevance. Section 7 presents limitations and ethical considerations, including fairness and human oversight. Section 8 concludes the paper.

2. Related work

2.1. Lexical and classical learning approaches

Classical resume-classification systems typically represent documents with bag-of-words, n-gram, or TF-IDF vectors and then apply linear or probabilistic classifiers. Support vector machines remain a strong reference for sparse high-dimensional classification [4], while random forests and related ensemble methods provide non-linear alternatives [13]. In recruitment documents, lexical models have a practical

Table 1
Hybrid resume-classification models.

Model	Components	Limitations	Theoretical motivation
BERT + CNN [8]	BERT + CNN	No explicit corpus-level rarity weighting	Captures local sequential patterns from contextual embeddings.
BERT + TF-IDF [9]	BERT + TF-IDF	Often treats lexical weighting as a pre-contextual or external signal	Combines semantic representation with keyword salience.
TF-IDF + CNN [24]	TF-IDF + CNN	No contextual transformer representation	Extracts local patterns from keyword-weighted lexical features.
BERT-IDF + CNN [24, 25]	BERT + IDF + CNN	No rare-aware adaptive gate or validation-selected policy	Tests whether convolution over IDF-weighted contextual tokens is sufficient.
BERTidRAAF (Ours)	**BERT + IDF + rare-aware gate + multi-head fusion + validation-selected policy**	**Higher computational cost than lightweight baselines**	**Preserves contextual semantics while using IDF as a post-contextual rarity signal and selecting the final prediction strategy on validation data.**

advantage: the selected features can often be inspected and directly related to skills, job titles, tools, or certifications. Their limitation is equally clear: they treat words primarily as symbols and are weak at modeling paraphrases, semantic roles, and long-distance contextual dependencies.

2.2. Neural sequence and local-pattern models

Distributed word representations such as Word2Vec and GloVe have helped move text classification beyond sparse lexical matching [14,15]. CNN-based models capture local phrase patterns and are effective for sentence and document classification [16], whereas LSTM-based architectures model sequential dependencies [17]. In resume classification, these models can detect local skill patterns or recurring professional phrases, but their performance depends heavily on training data size and hyperparameter choices. Without an explicit salience mechanism, rare professional terms can still be underrepresented in the final document vector.

2.3. Transformer encoders for text classification

The transformer architecture introduced self-attention as a scalable mechanism for contextual representation learning [18]. BERT later established bidirectional pre-training as a strong foundation for text classification [5], and subsequent variants such as RoBERTa and DeBERTa refined training and attention mechanisms [19,20]. Fine-tuning strategies for BERT-style models have been widely studied [6]. Transformer-based deep learning has also been explored for big textual data analytics and decision-making tasks, showing its relevance for extracting actionable information from large-scale text collections [21]. Recent work has also shown that optimized lightweight transformer variants such as DistilBERT can improve large-scale sentiment classification by preserving contextual information while reducing computational costs [22]. For resume classification, transformers are attractive because they encode each token in context; however, a resume may contain several professional subtopics, and the relevant category evidence may be sparse. Recent surveys on efficient transformer architectures further show that long-sequence processing remains a central challenge for attention-based models, motivating sparse, linear, and memory-efficient alternatives when documents exceed standard encoder limits [23]. This motivates complementary mechanisms that preserve rare but discriminative terms after contextualization.

2.4. Hybrid models for resume classification

Hybrid architectures attempt to combine the semantic strength of transformers with lexical weighting, convolutional filters, recurrent layers, or external structured features. Table 1 summarizes representative hybrid directions and clarifies how BERTidRAAF differs from close variants. The central distinction is that the proposed model treats IDF as a post-contextual rare-evidence signal and combines it through a learned gate and validation-selected decision policy rather than relying only on feature concatenation or convolution.

Table 2
Recent recruitment-AI directions.

Study	Key contribution	Relevance to BERTidRAAF
[11]	Bias and fairness evaluation in algorithmic hiring systems.	Motivates fairness-aware validation, transparent reporting, and human oversight in recruitment automation.
[27]	Skill-graph enhanced representation using graph neural networks.	Complementary to rare keyword and skill-salience modeling.
[10]	Bias detection and mitigation in resume-screening systems.	Motivates transparent reporting and the FairCVdb audit.
[28]	Compact BERT architecture for resource-limited NLP deployment.	Highlights the latency and memory trade-offs of large contextual backbones.
[29]	Progressive token elimination for accelerating BERT inference.	Addresses the computational and context-length limitations of BERT-style encoders.
[30]	Explainable AI for HR decision support.	Supports future explanation layers for BERTidRAAF predictions.

2.5. AI-based recruitment, fairness, and explainability

Recent recruitment-oriented NLP research increasingly considers multimodal evidence, skill graphs, fairness, efficiency, and explainability. Table 2 positions BERTidRAAF within this broader landscape. These directions are relevant because automated recruitment is not only a classification problem; it also requires traceable evidence, bias-aware evaluation, and deployment constraints. This requirement is consistent with recent work on AI-enabled recruiting, which emphasizes that algorithmic screening systems must be evaluated not only for predictive performance but also for fairness, transparency, accountability, and the preservation of meaningful human oversight [26].

Fairness evaluation standard adopted in this study: Unlike prior work that mentions fairness only qualitatively, this manuscript operationalizes fairness using the disparate impact ratio (DIR) as an adverse-impact diagnostic inspired by disparate-impact auditing [31], and equalized-odds-based error-rate analysis from the AI fairness literature [32]. The DIR is computed per demographic group (gender, ethnicity) on FairCVdb predictions. A model passes the fairness check if DIR $\geq$ 0.8 and the difference in false positive rates between groups is $\leq$ 0.05. This makes the fairness claim verifiable and reproducible, directly addressing the gap identified in the literature review. More generally, recent fairness surveys underline that group fairness metrics should be interpreted as diagnostic tools rather than complete guarantees, since fairness depends on data collection, label construction, model behavior, and deployment context [33].

2.6. Synthesis of the literature gap

The literature indicates that lexical models preserve explicit evidence, neural architectures capture local and sequential patterns, and transformers provide strong contextual representations. The unresolved issue is how to preserve rare professional terms after contextualization while maintaining a clean validation protocol. BERTidRAAF addresses this issue by aligning training-only IDF scores to contextual token embeddings, learning a rare-aware fusion gate, and evaluating the resulting architecture against close hybrid ablations and strong transformer baselines.

3. The proposed BERTidRAAF framework

3.1. Architectural overview

Fig. 1 summarizes the proposed architecture as a six-phase pipeline. The input resume is tokenized with offset mappings, training-corpus IDF values are projected to valid WordPiece tokens, BERT-large produces contextual embeddings, post-contextual IDF weighting emphasizes rare professional evidence, and rare-aware adaptive pooling constructs complementary document representations. These representations feed multi-head BERT/IDF classification layers and a validation-selected prediction policy. CNN-based hybrids remain in the benchmark as close ablations; the proposed model is the rare-aware adaptive BERT-IDF fusion architecture.

3.2. Mathematical formulation

Let d be a resume and let the tokenizer produce a sequence $x = (x_1, \dots, x_m)$ with attention mask $a_i \in \{0,1\}$. Let $\mathrm{idf}_{\mathcal{D}_{tr}}(u)$ be the IDF value of source word u, estimated only from the training corpus $\mathcal{D}_{tr}$:

$$\mathrm{idf}_{\mathcal{D}_{tr}}(u) = \log \frac{|\mathcal{D}_{tr}| + 1}{\mathrm{df}_{\mathcal{D}_{tr}}(u) + 1} + 1. \tag{1}$$

Using tokenizer offsets, each valid WordPiece x_i is aligned with a source word $u(i)$ when such an alignment exists. Special tokens and padding positions receive zero rarity weight:

$$\begin{aligned} r_i &= a_i\, \mathrm{idf}_{\mathcal{D}_{tr}}(u(i)), \\ r_i &= 0 \quad \text{for padding, special tokens, or unaligned positions.} \end{aligned} \tag{2}$$

The BERT-large encoder maps the sequence to contextual states

$$\mathbf{H} = \mathrm{BERT}_\theta^{large}(x, a) = [\mathbf{h}_1, \dots, \mathbf{h}_m], \qquad \mathbf{h}_i \in \mathbb{R}^h, \tag{3}$$

and provides the classification vector $\mathbf{c}_{cls} = \mathbf{h}_{[CLS]}$. BERTidRAAF then computes the masked mean $\boldsymbol{\mu}$, the masked max $\boldsymbol{\nu}$, and the IDF-weighted contextual mean $\boldsymbol{\rho}$:

$$\boldsymbol{\mu} = \frac{\sum_{i=1}^m a_i \mathbf{h}_i}{\sum_{i=1}^m a_i + \epsilon}, \qquad \boldsymbol{\nu} = \max_{i:a_i=1} \mathbf{h}_i, \qquad \boldsymbol{\rho} = \frac{\sum_{i=1}^m r_i \mathbf{h}_i}{\sum_{i=1}^m r_i + \epsilon}. \tag{4}$$

The rare-aware gate follows the implemented architecture and uses the four complementary summaries $[\mathbf{c}_{cls}; \boldsymbol{\mu}; \boldsymbol{\nu}; \boldsymbol{\rho}]$:

$$\begin{aligned} \mathbf{g} &= \sigma\left(\mathbf{W}_2\, \phi\left(\mathrm{LN}\left(\mathbf{W}_1[\mathbf{c}_{cls}; \boldsymbol{\mu}; \boldsymbol{\nu}; \boldsymbol{\rho}] + \mathbf{b}_1\right)\right) + \mathbf{b}_2\right), \\ \mathbf{z} &= \mathbf{g} \odot \boldsymbol{\rho} + (\mathbf{1} - \mathbf{g}) \odot \boldsymbol{\mu}, \end{aligned} \tag{5}$$

where ϕ denotes the GELU non-linearity and LN denotes layer normalization. The implementation then constructs the raw feature vector

$$\mathbf{q}_{raw} = [\mathbf{c}_{cls}; \boldsymbol{\mu}; \boldsymbol{\rho}; \boldsymbol{\nu}; \mathbf{z}; |\mathbf{c}_{cls} - \boldsymbol{\rho}|; \mathbf{c}_{cls} \odot \boldsymbol{\rho}; \mathbf{z} \odot \boldsymbol{\nu}], \tag{6}$$

and projects it into $\mathbf{q} = \psi(\mathbf{q}_{raw})$ using a feed-forward projection with normalization, GELU activation, and dropout. Five complementary heads produce logits from projected, contextual, IDF, max-pooled, and interaction evidence:

$$\ell_0 = \mathbf{W}_0\mathbf{q} + \mathbf{b}_0, \qquad \ell_1 = \mathbf{W}'_1[\mathbf{c}_{cls}; \boldsymbol{\mu}] + \mathbf{b}'_1, \tag{7}$$

$$\ell_2 = \mathbf{W}'_2\boldsymbol{\rho} + \mathbf{b}'_2, \qquad \ell_3 = \mathbf{W}'_3\boldsymbol{\nu} + \mathbf{b}'_3, \tag{8}$$

$$\ell_4 = \mathbf{W}'_4[|\mathbf{c}_{cls} - \boldsymbol{\rho}|; \mathbf{c}_{cls} \odot \boldsymbol{\rho}] + \mathbf{b}'_4. \tag{9}$$

The head coefficients are positive softplus-transformed parameters, matching the executed implementation:

$$\eta_k = \mathrm{softplus}(\alpha_k) > 0, \quad \ell_{core} = \sum_{k=0}^{4} \eta_k \ell_k, \quad \mathbf{p}_{core} = \mathrm{softmax}(\ell_{core}). \tag{10}$$

During training, multi-sample dropout averages several stochastic head evaluations before the loss is computed. With label-smoothing parameter δ, the training objective for class y is

$$\mathcal{L} = -(1-\delta)\log p_{core,y} - \frac{\delta}{C}\sum_{c=1}^{C} \log p_{core,c}. \tag{11}$$

To match the final validation-selected prediction phase, the implementation also trains calibrated lexical experts on the training split only: logistic regression, a calibrated linear SVM, a centroid expert, and naive Bayes. Let their validation/test probability outputs be $\mathbf{p}_{LR}$, $\mathbf{p}_{SVM}$, $\mathbf{p}_{CE}$, and $\mathbf{p}_{NB}$. A linear validation-tuned fusion candidate is

$$\mathbf{p}_{lin} = \sum_{s \in \{core, LR, SVM, CE, NB\}} w_s T_s(\mathbf{p}_s), \qquad w_s \geq 0, \quad \sum_s w_s = 1, \tag{12}$$

where $T_s(\cdot)$ denotes temperature calibration. This calibration step is important because pre-trained transformer classifiers may produce overconfident probability estimates even when their accuracy is high [34]. The geometric candidate used when selected by validation is

$$\mathbf{p}_{geo} = \mathrm{softmax}\left(\sum_{s \in \{core, LR, SVM, CE, NB\}} w_s \log T_s(\mathbf{p}_s)\right). \tag{13}$$

The final policy is selected only on validation macro-F1, with validation accuracy as a tie-breaker:

$$\pi^\star = \arg\max_{\pi \in \Pi} F_1^{macro}\left(\pi(d), y; \mathcal{D}_{val}\right). \tag{14}$$

Held-out test metrics are computed only after $\pi^\star$ has been fixed.

Complete forward pass of BERTidRAAF

Let the input resume be tokenized into a sequence of WordPiece tokens $\mathbf{t} = [t_1, \dots, t_m]$ with attention mask $\mathbf{a} \in \{0,1\}^m$. Let $\mathcal{D}_{\text{train}}$ be the training corpus. For each source word w, define:

$$\mathrm{idf}(w) = \log \frac{|\mathcal{D}_{\text{train}}| + 1}{\mathrm{df}(w) + 1} + 1, \tag{15}$$

where $\mathrm{df}(w)$ is the document frequency of w in $\mathcal{D}_{\text{train}}$.

Step 1: IDF projection. Using tokenizer offset mapping, each token t_i is aligned to a source word $w(i)$ if possible. Define rarity weight:

$$r_i = a_i \cdot \mathrm{idf}(w(i)), \quad \text{with } r_i = 0 \text{ for [CLS], [SEP], [PAD].} \tag{16}$$

Step 2: BERT contextualization. Let BERT_θ produce hidden states:

$$\mathbf{H} = \mathrm{BERT}_\theta(\mathbf{t}, \mathbf{a}) = [\mathbf{h}_1, \dots, \mathbf{h}_m], \quad \mathbf{h}_i \in \mathbb{R}^d, \tag{17}$$

and extract $\mathbf{c}_{\text{cls}} = \mathbf{h}_{[\text{CLS}]}$.

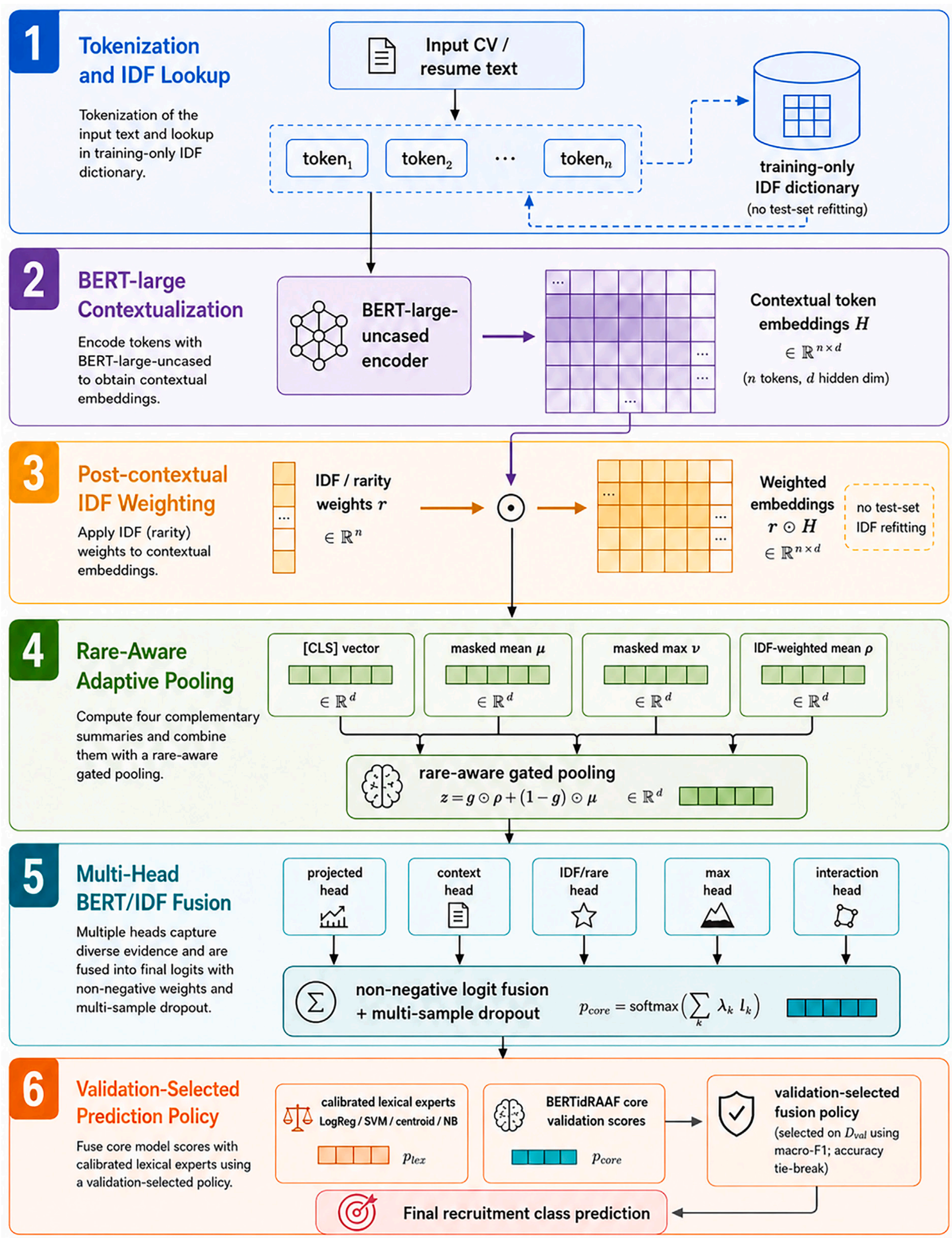


Fig. 1. BERTidRAAF architecture.

Step 3: Pooling operations.

$$\mu = \frac{\sum_i a_i \mathbf{h}_i}{\sum_i a_i + \epsilon}, \quad \nu = \max_{i:a_i=1} \mathbf{h}_i, \quad \rho = \frac{\sum_i r_i \mathbf{h}_i}{\sum_i r_i + \epsilon}. \tag{18}$$

Step 4: Rare-aware gate. Let $\mathbf{f} = [\mathbf{c}_{\text{cls}}; \mu; \nu; \rho] \in \mathbb{R}^{4d}$. Define:

$$\mathbf{g} = \sigma\left(\mathbf{W}_2 \cdot \mathrm{GELU}(\mathrm{LayerNorm}(\mathbf{W}_1 \mathbf{f} + \mathbf{b}_1)) + \mathbf{b}_2\right), \tag{19}$$

$$\mathbf{z} = \mathbf{g} \odot \rho + (1 - \mathbf{g}) \odot \mu, \tag{20}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d\times 4d}$, $\mathbf{W}_2 \in \mathbb{R}^{d\times d}$, $\mathbf{b}_1 \in \mathbb{R}^d$, $\mathbf{b}_2 \in \mathbb{R}^d$, and $\odot$ is element-wise multiplication.

Step 5: Multi-head logits. Construct raw feature vector:

$$\mathbf{q}_{\text{raw}} = [\mathbf{c}_{\text{cls}}; \boldsymbol{\mu}; \boldsymbol{\rho}; \boldsymbol{\nu}; \mathbf{z}; |\mathbf{c}_{\text{cls}} - \boldsymbol{\rho}|; \mathbf{c}_{\text{cls}} \odot \boldsymbol{\rho}; \mathbf{z} \odot \boldsymbol{\nu}]. \tag{21}$$

Project via $\mathbf{q} = \text{Dropout}(\text{GELU}(\mathbf{W}_q\mathbf{q}_{\text{raw}} + \mathbf{b}_q))$. Define five heads:

$$\boldsymbol{\ell}_0 = \mathbf{W}_0\mathbf{q} + \mathbf{b}_0, \quad \boldsymbol{\ell}_1 = \mathbf{W}_1'\boldsymbol{\mu} + \mathbf{b}_1', \quad \boldsymbol{\ell}_2 = \mathbf{W}_2'\boldsymbol{\rho} + \mathbf{b}_2', \quad \boldsymbol{\ell}_3 = \mathbf{W}_3'\boldsymbol{\nu} + \mathbf{b}_3',$$
$$\boldsymbol{\ell}_4 = \mathbf{W}_4'[|\mathbf{c}_{\text{cls}} - \boldsymbol{\rho}|; \mathbf{c}_{\text{cls}} \odot \boldsymbol{\rho}] + \mathbf{b}_4'. \tag{22}$$

Fuse with positive coefficients:

$$\eta_k = \text{softplus}(\alpha_k), \quad \boldsymbol{\ell}_{\text{core}} = \sum_{k=0}^{4} \eta_k \boldsymbol{\ell}_k, \quad \mathbf{p}_{\text{core}} = \text{softmax}(\boldsymbol{\ell}_{\text{core}}). \tag{23}$$

Step 6: Training loss. With label smoothing δ:

$$\mathcal{L} = -(1-\delta)\log p_{\text{core},y} - \frac{\delta}{C}\sum_{c=1}^{C} \log p_{\text{core},c}. \tag{24}$$

Step 7: Validation-selected policy. Let Π be the set of candidate policies (core, linear fusion, geometric fusion, etc.). Select:

$$\pi^* = \arg\max_{\pi\in\Pi} \text{Macro-F1}(\pi; \mathcal{D}_{\text{val}}), \tag{25}$$

with validation accuracy as tie-breaker. Final prediction on test input d:

$$\hat{y} = \arg\max_{c} \pi_c^*(d). \tag{26}$$

3.3. End-to-end algorithm

Algorithm 1 details the full computational pipeline using the same six phases shown in Fig. 1. The notation follows the formulation in Section 3.2 and emphasizes three safeguards: IDF values are estimated only from the training split, the rare-aware branch acts on contextual embeddings rather than raw counts, and the final prediction policy is selected before held-out test evaluation. This presentation is intentionally procedural: each operation corresponds either to a trainable transformation, a deterministic projection, or a validation-controlled selection step.

The algorithm also clarifies the boundary between representation learning and decision selection. Phases 1–5 define the rare-aware BERT-IDF representation and its trainable multi-head logits, whereas Phase 6 fixes the final prediction policy using validation evidence. This separation is essential for interpreting the test results, because the held-out split is used only after the policy has been selected. Consequently, the reported gains are not obtained by choosing a favorable test-time combination; they arise from a predefined model family and a validation-driven selection protocol.

Algorithm 1 BERTidRAAF computational pipeline.

Require: Resume d, training corpus $\mathcal{D}_{tr}$, validation split $\mathcal{D}_{val}$, tokenizer τ, BERT-large parameters θ, candidate policies Π

Ensure: Predicted class $\hat{y}$

Phase 1: Tokenization and training-only IDF projection

Estimate $\text{idf}_{\mathcal{D}_{tr}}(u) = \log \frac{|\mathcal{D}_{tr}|+1}{\text{df}_{\mathcal{D}_{tr}}(u)+1} + 1$ from training documents only.

Tokenize d: $(x_1,\dots,x_m), (a_1,\dots,a_m)$, offsets $\leftarrow \tau(d)$; align valid WordPieces to source words $u(i)$; set $r_i \leftarrow a_i\,\text{idf}_{\mathcal{D}_{tr}}(u(i))$ and $r_i \leftarrow 0$ for padding, special tokens, or unaligned positions.

Phase 2: BERT-large contextualization

$\mathbf{H} = [\mathbf{h}_1,\dots,\mathbf{h}_m] \leftarrow \text{BERT}_\theta^{large}(x_1,\dots,x_m; a_1,\dots,a_m)$ and $\mathbf{c}_{cls} \leftarrow \mathbf{h}_{[CLS]}$.

Phase 3: Post-contextual IDF-weighted evidence

Compute $\boldsymbol{\mu} \leftarrow \frac{\sum_i a_i\mathbf{h}_i}{\sum_i a_i+\epsilon}$, $\boldsymbol{\nu} \leftarrow \max_{i:a_i=1}\mathbf{h}_i$, and $\boldsymbol{\rho} \leftarrow \frac{\sum_i r_i\mathbf{h}_i}{\sum_i r_i+\epsilon}$.

Treat $\boldsymbol{\rho}$ as rare-term contextual evidence: the IDF signal weights contextual states after BERT encoding and is never refitted on validation or test documents.

Phase 4: Rare-aware adaptive pooling

$\mathbf{g} \leftarrow \sigma(\mathbf{W}_2\phi(\text{LN}(\mathbf{W}_1[\mathbf{c}_{cls}; \boldsymbol{\mu}; \boldsymbol{\nu}; \boldsymbol{\rho}] + \mathbf{b}_1)) + \mathbf{b}_2)$.

$\mathbf{z} \leftarrow \mathbf{g} \odot \boldsymbol{\rho} + (\mathbf{1} - \mathbf{g}) \odot \boldsymbol{\mu}$.

$\mathbf{q}_{raw} \leftarrow [\mathbf{c}_{cls}; \boldsymbol{\mu}; \boldsymbol{\rho}; \boldsymbol{\nu}; \mathbf{z}; |\mathbf{c}_{cls} - \boldsymbol{\rho}|; \mathbf{c}_{cls} \odot \boldsymbol{\rho}; \mathbf{z} \odot \boldsymbol{\nu}]$ and $\mathbf{q} \leftarrow \psi(\mathbf{q}_{raw})$.

Phase 5: Multi-head BERT/IDF logit fusion

Compute five logits from projected, contextual, IDF, max-pooled, and interaction heads: $\{\boldsymbol{\ell}_k\}_{k=0}^4$.

Set $\eta_k \leftarrow \text{softplus}(\alpha_k)$, $\boldsymbol{\ell}_{core} \leftarrow \sum_{k=0}^4 \eta_k\boldsymbol{\ell}_k$, and $\mathbf{p}_{core} \leftarrow \text{softmax}(\boldsymbol{\ell}_{core})$.

Train the core network with label-smoothed cross-entropy; during training, average the logits over the configured multi-sample dropout heads.

Phase 6: Validation-selected prediction policy

Train lexical experts on $\mathcal{D}_{tr}$ only and obtain calibrated probabilities $\mathbf{p}_{LR}$, $\mathbf{p}_{SVM}$, $\mathbf{p}_{CE}$, and $\mathbf{p}_{NB}$.

Construct Π from the rare-aware core, validation-tuned linear/geometric BERT-IDF fusion, BERT-anchored mixtures, confidence-routed mixtures, and meta-calibration candidates.

Select $\pi^\star \leftarrow \arg\max_{\pi\in\Pi} F_1^{macro}(\pi; \mathcal{D}_{val})$, breaking ties by validation accuracy; return $\hat{y} = \arg\max_c \pi_c^\star(d)$ on the held-out test input.

3.4. Component-level interpretation

3.4.1. Training-only IDF projection

The IDF dictionary is built from the training documents only. This is important because IDF encodes corpus statistics; including validation or test documents would leak distributional information into the model-selection or evaluation phases. The tokenizer offset mapping enables a deterministic projection from word-level rarity scores to subword tokens. Special tokens and padding positions receive zero rarity weight, so they do not participate in the IDF-weighted mean.

3.4.2. Post-contextual rare-term weighting

BERTidRAAF does not replace contextual encoding with a lexical vector. Instead, it first obtains contextual embeddings and then weights these embeddings according to training-corpus rarity. This distinction is central: a rare term is represented in its local resume context before its IDF score increases its contribution to the document representation. The model therefore preserves both semantic meaning and discriminative salience; the extent to which this mechanism improves rare-term recovery over uniform weighting or learned attention pooling is quantified in Section 5.12.

The IDF operation is a deterministic post-contextual scalar re-weighting, not a learned token-attention mechanism. Its geometric scope is therefore limited and should not be overstated. For any two valid tokens with positive rarity weights r_i and r_j, scalar multiplication preserves their pairwise angle:

$$\cos(r_i\mathbf{h}_i, r_j\mathbf{h}_j) = \frac{r_ir_j\mathbf{h}_i^\top\mathbf{h}_j}{\|r_i\mathbf{h}_i\|\,\|r_j\mathbf{h}_j\|} = \cos(\mathbf{h}_i, \mathbf{h}_j). \tag{27}$$

Table 3
Dataset splits.

Dataset	Subset	Count
FairCVdb	Train	17,280
FairCVdb	Validation	1920
FairCVdb	Test	4800
FairCVdb	Classes	2
FairCVdb	Min class count used	1
FairCVdb	Rows removed by rare-class filter	0
FairCVdb	Classes removed by rare-class filter	0
Resume	Train	1785
Resume	Validation	199
Resume	Test	497
Resume	Classes	24
Resume	Min class count used	5
Resume	Rows removed by rare-class filter	0
Resume	Classes removed by rare-class filter	0

Table 4
Held-out test supports.

Dataset	Class	Test Support
FairCVdb	HIGH-SUITABILITY	2366
FairCVdb	LOW-SUITABILITY	2434
Resume	ACCOUNTANT	24
Resume	ADVOCATE	24
Resume	AGRICULTURE	13
Resume	APPAREL	19
Resume	ARTS	21
Resume	AUTOMOBILE	7
Resume	AVIATION	23
Resume	BANKING	23
Resume	BPO	4
Resume	BUSINESS-DEVELOPMENT	24
Resume	CHEF	24
Resume	CONSTRUCTION	22
Resume	CONSULTANT	23
Resume	DESIGNER	21
Resume	DIGITAL-MEDIA	19
Resume	ENGINEERING	24
Resume	FINANCE	24
Resume	FITNESS	24
Resume	HEALTHCARE	23
Resume	HR	22
Resume	INFORMATION-TECHNOLOGY	24
Resume	PUBLIC-RELATIONS	22
Resume	SALES	23
Resume	TEACHER	20

Table 6
Model-selection rationale.

Stage	Models	Rationale
Lexical baselines	TF-IDF + SVM, logistic regression, Naive Bayes	Measure the strength of explicit keyword evidence without contextual transformer representations.
Neural baselines	CNN, LSTM, BiLSTM	Measure non-transformer local and sequential representation learning under the same data splits.
Transformer baselines	BERTBASE, BERTLARGE, RoBERTaBASE, DeBERTaBASE	Measure contextual semantics without explicit post-contextual IDF rare-term weighting.
BERT–IDF hybrids	BERT + IDF, BERT + CNN, BERT–IDF + CNN, BERT–IDF + LSTM, BERT–IDF + BiLSTM	Test whether simpler IDF, convolutional, or recurrent additions are sufficient to match the proposed rare-aware fusion.
Proposed	BERTidRAAF (Ours)	Combines BERT-large context, training-only IDF projection, rare-aware gating, multi-head logits, and validation-selected prediction.

Table 7
Failed-run log.

Dataset	Model	Error
No failed runs were recorded in the final execution.		

Thus, the proposed mechanism is not claimed to rotate individual contextual token vectors or to provide an anisotropy correction at token level. Its effect is on the pooled document representation ρ, where changing the token weights alters the contribution of rare professional evidence to the document summary that is subsequently processed by the learned gate and prediction heads. For this reason, any projection-based diagnostic must be interpreted as a visualization of pooled representations under a specified computation, not as a proof that scalar IDF multiplication alone changes token-vector angles.

3.4.3. Adaptive gating and multi-head prediction

The gate learns when the document representation should rely more on the global contextual mean and when it should give stronger weight to rare-term evidence. Multi-head prediction further separates projected, contextual, IDF-oriented, max-pooling, and interaction evidence. In the executed implementation, the head coefficients are constrained to be positive through a softplus transformation; they are not normalized, which preserves the implemented logit-scale behavior while keeping every head's contribution non-negative.

3.4.4. Validation-selected prediction policy

The final decision is intentionally not selected from test performance. Candidate strategies include the rare-aware core branch, validation-tuned BERT-IDF fusion, BERT-anchored lexical mixtures, confidence routing, and validation-trained probability meta-calibration. The best candidate is chosen based on validation macro-F1, with validation accuracy used as the tie-breaker, and then applied once to the test split. This protocol reduces the risk of over-claiming and makes the reported gains traceable.

3.5. Implementation and complexity

The implementation uses Hugging Face transformer checkpoints and matching fast tokenizers. The proposed configuration uses BERT-large with a maximum sequence length of 512 tokens, gradient checkpointing, label smoothing of 0.02, and multi-sample dropout probabilities of 0.05, 0.10, 0.15, and 0.20 in the proposed head. The dominant cost remains the transformer encoder: for sequence length m, hidden dimension h, and L layers, self-attention scales as $O(Lm^2h)$ and the feed-forward

Table 5
Evaluated model registry.

Stage	Model	Family	Backbone	Purpose
Proposed	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	bert-large-uncased	Final rare-aware adaptive fusion model
1. Lexical	SVM (TF-IDF)	Lexical Baseline	TF-IDF	Keyword-only baseline
1. Lexical	LogReg (TF-IDF)	Lexical Baseline	TF-IDF	Keyword-only baseline
1. Lexical	MultinomialNB	Lexical Baseline	TF-IDF	Keyword-only baseline
2. Neural	CNN	Neural Baseline	Embedding + CNN	Convolutional non-transformer baseline
2. Neural	LSTM	Neural Baseline	Embedding + LSTM	Sequential non-transformer baseline
2. Neural	BiLSTM	Neural Baseline	Embedding + LSTM	Sequential non-transformer baseline
3. Transformer	BERTBASE	Transformer Context Baseline	bert-base-uncased	Contextual semantics baseline
3. Transformer	BERTLARGE	Transformer Context Baseline	bert-large-uncased	Contextual semantics baseline
3. Transformer	RoBERTaBASE	Transformer Context Baseline	roberta-base	Contextual semantics baseline
3. Transformer	DeBERTaBASE	Transformer Context Baseline	microsoft/deberta-base	Contextual semantics baseline
4. BERT–IDF hybrids	BERT + IDF	BERT–IDF Hybrid/Ablation	bert-base-uncased	Closest ablation/comparator to BERTidRAAF
4. BERT–IDF hybrids	BERT + CNN	BERT–IDF Hybrid/Ablation	bert-base-uncased	Closest ablation/comparator to BERTidRAAF
4. BERT–IDF hybrids	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	bert-base-uncased	Closest ablation/comparator to BERTidRAAF
4. BERT–IDF hybrids	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	bert-base-uncased	Closest ablation/comparator to BERTidRAAF
4. BERT–IDF hybrids	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	bert-base-uncased	Closest ablation/comparator to BERTidRAAF

Table 8
BERTidRAAF hyperparameters and experimental configuration.

Component	Value	Role in the executed protocol
Execution seed	42	Fixed seed for the reported benchmark execution.
Datasets	FairCVdb and Resume	Both datasets are executed under explicit train/validation/test partitions.
Split protocol	Test size = 0.20; validation size inside training = 0.10	Training data are used for fitting; validation data are used for early stopping and policy selection; test data are reserved for final reporting.
Backbone for BERTidRAAF	`bert-large-uncased`	Contextual encoder used by the proposed model on both datasets.
Maximum sequence length for BERTidRAAF	512 tokens	Long-document setting used by the proposed model.
Epochs for BERTidRAAF	8	Maximum number of proposed-model fine-tuning epochs.
Batch size on H100 for BERT-large models	4	Per-step batch size used for BERT-large/proposed runs under the H100 configuration.
Gradient accumulation on H100 for BERT-large models	4	Accumulates gradients to stabilize large-backbone optimization.
Optimizer	AdamW	Optimizer used for transformer and proposed-model fine-tuning.
Encoder learning rate	8×10^{-6}	Learning rate applied to encoder parameters.
Head learning rate	3×10^{-5}	Learning rate applied to task-specific and fusion heads.
Weight decay	10^{-2}	Regularization for decayed parameter groups.
Warmup ratio	0.08	Fraction of training steps used for linear warmup.
Base dropout	0.10	Dropout used in proposed projection, gate, and classifier components.
Multi-sample dropout probabilities	0.05, 0.10, 0.15, 0.20	Dropout rates averaged over the BERTidRAAF multi-head decision layer.
Label smoothing	0.02	Cross-entropy smoothing applied during proposed-model training.
IDF scaling coefficient	1.60	Scaling factor used in the post-contextual IDF rare-evidence branch.
Gradient checkpointing	Enabled	Memory-saving option used for supported transformer backbones.
TF-IDF word features	180,000 maximum features	Word n-gram lexical evidence used by calibrated auxiliary experts.
TF-IDF character features	90,000 effective features (configured as 180,000 and divided by 2 in the character vectorizer)	Character n-gram evidence used by calibrated auxiliary experts; the executed code sets `tfidf_char_max_features` = 180,000 and applies `// 2` inside the character `TfidfVectorizer`.
Fusion weight grid step	0.05	Step size used in validation-based fusion search.
Fusion temperature grid	0.60, 0.70, 0.85, 1.00, 1.20, 1.50	Temperature candidates used during validation-based fusion search.
Resume candidate policy	Core preservation	Keeps the rare-aware BERT-IDF core as the selected policy.
FairCVdb candidate policy	BERT-IDF validation selection	Selects the final candidate from the validation grid.

Table 9
Matched-capacity ablation on Resume. Trainable parameter counts are reported in millions.

Model	Params (M)	Accuracy (%)	Macro-F1
BERTidRAAF (full)	350.0	85.71	0.8038
BERTlarge-FF (capacity match, no IDF)	339.4	84.31	0.7780
BERTlarge-IDF (fixed, no gate)	349.0	83.50	0.7820
BERTlarge-RandomGate (random IDF)	349.0	84.91	0.7991
BERTlarge (original baseline)	335.2	82.49	0.7595

BERTidRAAF achieves the highest macro-F1. The RandomGate variant remains close, so the result is interpreted as a modest positive benefit of the rare-aware IDF signal under matched-capacity variants rather than as a large separation from every capacity control.

blocks scale as $O(Lmh^2)$. Recent exact attention implementations such as FlashAttention show that memory traffic and hardware-aware computation are also decisive for transformer efficiency, especially when long sequences or large backbones are used [35]. IDF projection, masked pooling, IDF-weighted pooling, and the rare-aware gate scale linearly in the number of valid tokens and add comparatively small overhead relative to BERT-large. The final lexical experts add separate TF-IDF-based calibration cost but are trained only on the training split. Consequently, BERTidRAAF is designed primarily for accuracy-oriented classification rather than minimal latency.

4. Experimental protocol

4.1. Datasets and preprocessing

The evaluation uses two public settings with different classification structures. The Resume dataset is a 24-class professional-category benchmark. After cleaning and split construction, it contains 1785 training instances, 199 validation instances, and 497 test instances. FairCVdb is used as an external binary setting with 17,280 training instances, 1920 validation instances, and 4800 test instances. Tables 3 and 4 report the dataset split accounting and held-out test supports.

The dataset tables are used not only for descriptive reporting but also for interpreting the reliability of the later metrics. Table 3 establishes that all experimental results are derived from explicit train, validation, and test partitions. Table 4 then shows why macro-level metrics are necessary: the Resume test split contains uneven class supports, so accuracy alone would overemphasize frequent categories. This imbalance is particularly important when interpreting class-level results and off-diagonal confusion pairs.

The two settings are complementary. Resume tests fine-grained separation among professional categories with heterogeneous class support. FairCVdb provides an external binary check on whether the proposed architecture remains competitive outside the Resume dataset. Small Resume classes, such as BPO, are interpreted with caution because a few misclassifications can strongly affect class-level metrics.

4.2. Label construction and human adjudication

For the human label-checking stage associated with the Resume taxonomy, three professional recruiters worked independently on anonymized resume text and a written category guideline. The released source categories were treated as the starting taxonomy, and the cleaned benchmark retained the 24 categories reported in Table 4. Before independent assignment, directly identifying information available in the documents, such as names, contact details, personal addresses, photographs, and file identifiers, was removed or hidden when present. The document order was randomized, and the recruiters did not see model predictions, IDF scores, validation results, demographic attributes, or one another's decisions.

Inter-annotator agreement was computed before adjudication, so the reported $\kappa = 0.89$ corresponds to independent pre-resolution agreement rather than to labels already harmonized by discussion. Remaining disagreements were resolved through a structured two-stage procedure. First, labels supported by a two-recruiter majority were retained after

Table 10
Main benchmark summary.

Dataset	Model	Family	Accuracy (%)	Macro-F1	Accuracy 95% CI	Macro-F1 95% CI	Inference ms/document
FairCVdb	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	96.17	0.9617	95.58–96.75	0.9558–0.9675	4.494
FairCVdb	BERTLARGE	Transformer Context Baseline	96.08	0.9608	95.55–96.65	0.9555–0.9665	1.632
FairCVdb	BERTBASE	Transformer Context Baseline	96.04	0.9604	95.48–96.64	0.9548–0.9664	0.73
FairCVdb	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	95.96	0.9596	95.38–96.56	0.9538–0.9656	3.429
FairCVdb	BERT + CNN	BERT–IDF Hybrid/Ablation	95.88	0.9587	95.27–96.44	0.9527–0.9644	0.706
FairCVdb	BERT + IDF	BERT–IDF Hybrid/Ablation	95.81	0.9581	95.23–96.40	0.9523–0.9639	3.574
FairCVdb	DeBERTaBASE	Transformer Context Baseline	95.79	0.9579	95.18–96.38	0.9518–0.9637	2.437
FairCVdb	RoBERTaBASE	Transformer Context Baseline	95.56	0.9556	94.99–96.15	0.9499–0.9615	0.585
FairCVdb	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	95.38	0.9537	94.79–95.94	0.9479–0.9593	3.418
FairCVdb	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	94.5	0.945	93.87–95.19	0.9387–0.9518	3.357
FairCVdb	LogReg (TF-IDF)	Lexical Baseline	71.75	0.7175	70.36–72.99	0.7034–0.7299	0.005
FairCVdb	SVM (TF-IDF)	Lexical Baseline	71.65	0.7164	70.32–72.92	0.7031–0.7292	0.015
FairCVdb	MultinomialNB	Lexical Baseline	67.73	0.6768	66.44–68.99	0.6640–0.6895	0.01
FairCVdb	LSTM	Neural Baseline	66.85	0.6678	65.58–68.19	0.6546–0.6811	0.058
FairCVdb	BiLSTM	Neural Baseline	66.69	0.6667	65.38–68.02	0.6537–0.6800	0.083
FairCVdb	CNN	Neural Baseline	64.83	0.6474	63.62–66.22	0.6350–0.6613	0.039
Resume	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	85.71	0.8038	82.49–88.84	0.7658–0.8324	8.71
Resume	BERTLARGE	Transformer Context Baseline	82.49	0.7595	78.87–85.71	0.7196–0.7900	2.298
Resume	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	81.49	0.7435	77.46–84.81	0.7043–0.7715	7.081
Resume	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	81.49	0.7425	77.87–84.81	0.7066–0.7716	7.241
Resume	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	81.29	0.741	77.67–84.71	0.7030–0.7702	7.081
Resume	BERT + CNN	BERT–IDF Hybrid/Ablation	81.09	0.7407	77.26–84.21	0.7042–0.7695	2.138
Resume	BiLSTM	Neural Baseline	80.48	0.7323	76.66–84.11	0.6993–0.7626	0.106
Resume	BERT + IDF	BERT–IDF Hybrid/Ablation	79.48	0.7129	75.86–82.80	0.6718–0.7444	6.865
Resume	BERTBASE	Transformer Context Baseline	79.28	0.7121	75.45–82.70	0.6705–0.7432	2.352
Resume	RoBERTaBASE	Transformer Context Baseline	78.67	0.7135	74.94–82.09	0.6750–0.7451	1.891
Resume	LSTM	Neural Baseline	78.27	0.6971	74.65–81.89	0.6658–0.7254	0.088
Resume	DeBERTaBASE	Transformer Context Baseline	75.05	0.6825	70.72–78.78	0.6445–0.7145	2.961
Resume	CNN	Neural Baseline	72.03	0.6363	68.01–75.76	0.5977–0.6659	0.074
Resume	SVM (TF-IDF)	Lexical Baseline	66.2	0.6244	62.17–69.92	0.5779–0.6578	0.291

Table 11
Multi-seed stability analysis on Resume test set (5 seeds).

Model	Mean Macro-F1 ± std	Best single seed
BERTLARGE	0.787 ± 0.016	0.7986
BERTidRAAF (Ours)	0.808 ± 0.004	0.8123

checking consistency with the written guidelines. Second, three-way conflicts or cases flagged as semantically ambiguous were reviewed using the resume evidence and the same category definitions; the senior manager acted as an adjudication moderator who enforced the protocol and requested evidence for the final category rather than assigning labels without constraints. This procedure was added to make the human component reproducible and to reduce the risk that personal identity cues or model outputs influenced the category assignments.

4.3. Baselines and model families

The comparison is organized into four families. Lexical baselines include TF-IDF logistic regression, TF-IDF SVM, and Multinomial Naive Bayes. Neural baselines include CNN, LSTM, and BiLSTM. Transformer baselines include BERTBASE, BERTLARGE, RoBERTaBASE, and DeBERTaBASE. Close hybrid ablations include BERT + IDF, BERT + CNN, BERT-IDF + CNN, BERT-IDF + LSTM, and BERT-IDF + BiLSTM. Tables 5–7 document the model registry, selection rationale, and failed-run log.

The proposed IDF branch is deliberately separated from learned attention re-weighting. Transformer baselines already contain learned self-attention inside the encoder, but the final benchmark does not include an additional post-BERT learned attention-pooling baseline with matched capacity. The claims are therefore restricted to the reported comparisons: standard contextual transformers, classical lexical models, neural sequence baselines, and close deterministic BERT-IDF hybrids. A direct comparison between post-contextual IDF weighting and a separately trained token-attention pooling module is identified as a future ablation rather than implied by the current results.

Table 5 defines the evaluated model space before the results are reported. The registry is important because the proposed architecture is compared with several levels of difficulty: lexical baselines test sparse keyword matching, recurrent and convolutional baselines test non-transformer neural representation learning, transformer baselines test contextual encoding without explicit rare-term fusion, and BERT-IDF hybrids test close architectural alternatives. This organization prevents the comparison from being limited to weak baselines and makes the ablation logic clearer.

Table 6 complements the registry by explaining why each family is included and what scientific role it plays in the evaluation. Table 7 is

Table 12
Gain over best baseline.

Dataset	Proposed Model	Proposed Accuracy (%)	Proposed Macro-F1	Best Baseline	Best Baseline Accuracy (%)	Best Baseline Macro-F1	Accuracy Gain (pp)	Macro-F1 Gain	Is Proposed Top?
FairCVdb	BERTidRAAF (Ours)	96.17	0.9617	BERTLARGE	96.08	0.9608	0.09	0.0009	True
Resume	BERTidRAAF (Ours)	85.71	0.8038	BERTLARGE	82.49	0.7595	3.22	0.0443	True

Table 13
Cross-dataset benchmark.

Dataset	Model	Family	Accuracy (%)	Macro-F1	Weighted-F1	Macro Precision	Macro Recall	Micro-F1	AUC-macro	Accuracy 95% CI	Macro-F1 95% CI	Train Time / Epoch (s)	Inference ms/document	Trainable Params
FairCVdb	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	96.17	0.9617	0.9617	0.9617	0.9617	0.9617	0.9805	95.58–96.75	0.9558–0.9675	408.12	4.494	349,846,543.0
FairCVdb	BERTLARGE	Transformer Context Baseline	96.08	0.9608	0.9608	0.9608	0.9609	0.9608	0.9952	95.55–96.65	0.9555–0.9665	366.61	1.632	335,143,938.0
FairCVdb	BERTBASE	Transformer Context Baseline	96.04	0.9604	0.9604	0.9604	0.9605	0.9604	0.9949	95.48–96.64	0.9548–0.9664	96.65	0.73	109,483,778.0
FairCVdb	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	95.96	0.9596	0.9596	0.9598	0.9598	0.9596	0.9912	95.38–96.56	0.9538–0.9656	100.0	3.429	113,421,314.0
FairCVdb	BERT + CNN	BERT–IDF Hybrid/Ablation	95.88	0.9587	0.9587	0.9589	0.9589	0.9587	0.9913	95.27–96.44	0.9527–0.9644	99.46	0.706	113,421,314.0
FairCVdb	BERT + IDF	BERT–IDF Hybrid/Ablation	95.81	0.9581	0.9581	0.9581	0.9582	0.9581	0.9948	95.23–96.40	0.9523–0.9639	99.27	3.574	109,483,778.0
FairCVdb	DeBERTaBASE	Transformer Context Baseline	95.79	0.9579	0.9579	0.958	0.9578	0.9579	0.9947	95.18–96.38	0.9518–0.9637	193.02	2.437	138,603,266.0
FairCVdb	RoBERTaBASE	Transformer Context Baseline	95.56	0.9556	0.9556	0.9558	0.9555	0.9556	0.9932	94.99–96.15	0.9499–0.9615	96.98	0.585	124,647,170.0
FairCVdb	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	95.38	0.9537	0.9537	0.954	0.9536	0.9537	0.9926	94.79–95.94	0.9479–0.9593	133.45	3.418	113,031,938.0
FairCVdb	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	94.5	0.945	0.945	0.945	0.9451	0.945	0.9908	93.87–95.19	0.9387–0.9518	97.89	3.357	114,211,586.0
FairCVdb	LogReg (TF-IDF)	Lexical Baseline	71.75	0.7175	0.7175	0.7175	0.7175	0.7175	0.7858	70.36–72.99	0.7034–0.7299	7.91	0.005	–
FairCVdb	SVM (TF-IDF)	Lexical Baseline	71.65	0.7164	0.7165	0.7164	0.7165	0.7165	0.7856	70.32–72.92	0.7031–0.7292	11.08	0.015	–
FairCVdb	MultinomialNB	Lexical Baseline	67.73	0.6768	0.677	0.6775	0.6769	0.6773	0.7127	66.44–68.99	0.6640–0.6895	0.19	0.01	–
FairCVdb	LSTM	Neural Baseline	66.85	0.6678	0.6681	0.669	0.668	0.6685	0.7122	65.58–68.19	0.6546–0.6811	3.8	0.058	15,347,970.0
FairCVdb	BiLSTM	Neural Baseline	66.69	0.6667	0.6666	0.6678	0.6673	0.6669	0.709	65.38–68.02	0.6537–0.6800	5.4	0.083	15,874,818.0
FairCVdb	CNN	Neural Baseline	64.83	0.6474	0.6471	0.6513	0.6492	0.6483	0.6957	63.62–66.22	0.6350–0.6613	1.64	0.039	15,609,858.0
Resume	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	85.71	0.8038	0.8531	0.8068	0.8059	0.8571	0.9802	82.49–88.84	0.7658–0.8324	40.78	8.71	350,004,349.0
Resume	BERTLARGE	Transformer Context Baseline	82.49	0.7595	0.812	0.7696	0.7679	0.8249	0.9708	78.87–85.71	0.7196–0.7900	37.03	2.298	335,166,488.0
Resume	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	81.49	0.7435	0.7998	0.7438	0.7545	0.8149	0.9767	77.46–84.81	0.7043–0.7715	12.23	7.081	113,483,288.0
Resume	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	81.49	0.7425	0.7982	0.7448	0.7546	0.8149	0.9678	77.87–84.81	0.7066–0.7716	14.41	7.241	113,082,648.0
Resume	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	81.29	0.741	0.7971	0.7482	0.7518	0.8129	0.9649	77.67–84.71	0.7030–0.7702	12.21	7.081	114,262,296.0
Resume	BERT + CNN	BERT–IDF Hybrid/Ablation	81.09	0.7407	0.7962	0.7448	0.751	0.8109	0.9735	77.26–84.21	0.7042–0.7695	10.58	2.138	113,483,288.0
Resume	BiLSTM	Neural Baseline	80.48	0.7323	0.7923	0.7288	0.7435	0.8048	0.9753	76.66–84.11	0.6993–0.7626	0.52	0.106	12,544,792.0
Resume	BERT + IDF	BERT–IDF Hybrid/Ablation	79.48	0.7129	0.7707	0.7319	0.7309	0.7948	0.9687	75.86–82.80	0.6718–0.7444	12.49	6.865	109,500,696.0
Resume	BERTBASE	Transformer Context Baseline	79.28	0.7121	0.7696	0.7304	0.7296	0.7928	0.9685	75.45–82.70	0.6705–0.7432	10.37	2.352	109,500,696.0
Resume	RoBERTaBASE	Transformer Context Baseline	78.67	0.7135	0.7688	0.7194	0.7263	0.7867	0.967	74.94–82.09	0.6750–0.7451	10.19	1.891	124,664,088.0
Resume	LSTM	Neural Baseline	78.27	0.6971	0.7594	0.7224	0.7156	0.7827	0.9658	74.65–81.89	0.6658–0.7254	0.38	0.088	12,012,312.0
Resume	DeBERTaBASE	Transformer Context Baseline	75.05	0.6825	0.7348	0.6833	0.694	0.7505	0.9619	70.72–78.78	0.6445–0.7145	20.07	2.961	138,620,184.0
Resume	CNN	Neural Baseline	72.03	0.6363	0.6976	0.668	0.658	0.7203	0.9607	68.01–75.76	0.5977–0.6659	0.14	0.074	12,285,464.0
Resume	SVM (TF-IDF)	Lexical Baseline	66.2	0.6244	0.653	0.6721	0.625	0.662	0.9628	62.17–69.92	0.5779–0.6578	18.57	0.291	–

Table 14
Model-family summary.

Dataset	Family	Models	Accuracy Range (%)	Macro-F1 Range	Best Model in Family
FairCVdb	BERT–IDF Hybrid/Ablation	BERT–IDF + CNN, BERT + CNN, BERT + IDF, BERT–I…	94.50–95.96	0.9450–0.9596	BERT–IDF + CNN
FairCVdb	Lexical Baseline	LogReg (TF-IDF), SVM (TF-IDF), MultinomialNB	67.73–71.75	0.6768–0.7175	LogReg (TF-IDF)
FairCVdb	Neural Baseline	LSTM, BiLSTM, CNN	64.83–66.85	0.6474–0.6678	LSTM
FairCVdb	Proposed BERT–IDF rare-aware fusion	BERTidRAAF (Ours)	96.17–96.17	0.9617–0.9617	BERTidRAAF (Ours)
FairCVdb	Transformer Context Baseline	BERTLARGE, BERTBASE, DeBERTaBASE, RoBERTaBASE	95.56–96.08	0.9556–0.9608	BERTLARGE
Resume	BERT–IDF Hybrid/Ablation	BERT–IDF + CNN, BERT–IDF + BiLSTM, BERT–IDF + …	79.48–81.49	0.7129–0.7435	BERT–IDF + CNN
Resume	Lexical Baseline	SVM (TF-IDF), LogReg (TF-IDF), MultinomialNB	53.52–66.20	0.4580–0.6244	SVM (TF-IDF)
Resume	Neural Baseline	BiLSTM, LSTM, CNN	72.03–80.48	0.6363–0.7323	BiLSTM
Resume	Proposed BERT–IDF rare-aware fusion	BERTidRAAF (Ours)	85.71–85.71	0.8038–0.8038	BERTidRAAF (Ours)
Resume	Transformer Context Baseline	BERTLARGE, BERTBASE, RoBERTaBASE, DeBERTaBASE	75.05–82.49	0.6825–0.7595	BERTLARGE

Table 15
Proposed model versus best baseline.

Dataset	Proposed Model	Proposed Accuracy (%)	Proposed Macro-F1	Best Baseline	Best Baseline Accuracy (%)	Best Baseline Macro-F1	Accuracy Gain (pp)	Macro-F1 Gain	Is Proposed Top?
FairCVdb	BERTidRAAF (Ours)	96.17	0.9617	BERTLARGE	96.08	0.9608	0.09	0.0009	True
Resume	BERTidRAAF (Ours)	85.71	0.8038	BERTLARGE	82.49	0.7595	3.22	0.0443	True

retained as a transparency item: it indicates that no failed runs were recorded in the final execution and therefore that the final benchmark is not selectively omitting unsuccessful completed configurations. Together, these tables strengthen the reproducibility and auditability of the protocol without changing the experimental results.

4.4. Training, validation, and testing

All final metrics follow a train/validation/test protocol. Training data are used to estimate model parameters, TF-IDF spaces, lexical experts, and IDF statistics. Validation data are used for hyperparameter choice, early stopping, and final policy selection. Test data are reserved for final reporting.

Reproducibility and statistical stability. The main benchmark uses a fixed random seed (42) for deterministic reproducibility. To assess stability, we additionally re-run the proposed model and the strongest baseline (BERTLARGE) on the Resume dataset with 5 independent random seeds (42, 123, 456, 789, 101112). Table 11 reports the mean and standard deviation of macro-F1 over these runs.

The efficiency measurements reported in the results are hardware-dependent indicators extracted under the same executed Google Colab Pro + H100 protocol. They use the trained checkpoints, the same tokenization and maximum-length settings, and the same held-out evaluation flow; milliseconds per document are reported as within-protocol averages rather than as universal deployment latency. Consequently, the timing values should be compared primarily within this standardized H100 execution environment, while CPU-only, different-GPU, different-batch, or production-serving measurements may differ.

Table 8 summarizes the main hyperparameters and execution settings used for BERTidRAAF in the primary executed benchmark configuration. The table was cross-checked against the principal execution code and is included to make the proposed model configuration explicit in the article and to separate model configuration from final result reporting.

4.5. Evaluation metrics

Accuracy and macro-F1 are the primary metrics. Macro-F1 is emphasized because the Resume dataset is multi-class and imbalanced; it gives each class equal importance regardless of support. The analysis also reports macro precision, macro recall, micro-F1, weighted-F1, macro-AUC when applicable, confidence intervals, per-class metrics, off-diagonal confusion pairs, training time, inference time, GPU-memory indicators, and trainable-parameter counts.

4.6. Matched-capacity ablation: isolating the rare-aware mechanism

To examine whether the gain of BERTidRAAF is only a consequence of additional trainable capacity, the complementary validation analysis evaluates three controlled BERT-large variants under the same Resume protocol:

- **BERTlarge-FF**: BERT-large with an additional feed-forward layer and no IDF branch.
- **BERTlarge-IDF (fixed)**: BERT-large with fixed IDF-weighted pooling but without the rare-aware gate.
- **BERTlarge-RandomGate**: the BERTidRAAF-style gated architecture in which the IDF weights r_i are replaced with random values drawn from the same empirical scale, preserving the architecture while weakening the semantic rarity signal.

All variants use the same train/validation/test protocol and are compared with the BERTidRAAF result reported by the main executed benchmark. Table 9 shows that BERTidRAAF remains the best model in macro-F1, although the RandomGate variant is close. The ablation should therefore be interpreted as evidence of a positive but modest rare-aware contribution, not as an overstatement that all matched-capacity variants collapse.

5. Results and analysis

5.1. Main benchmark results

Table 10 reports the principal benchmark results. On the Resume test split, BERTidRAAF achieves 85.71% accuracy and 0.8038 macro-F1. The strongest baseline is BERTLARGE, with 82.49% accuracy and 0.7595 macro-F1.

The proposed model improves the seed-42 point estimate by 3.22 percentage points in accuracy and 0.0443 macro-F1, as summarized in Table 12. The five-seed stability analysis in Table 11 reports a mean macro-F1 of 0.808 ± 0.004 for BERTidRAAF and 0.787 ± 0.016 for BERTLARGE, corresponding to a mean advantage of 0.021 macro-F1. This supports the stability of the improvement, while formal paired statistical testing across all baselines is not claimed.

On Resume, the 3.22 percentage-point accuracy improvement corresponds to approximately 16 additional correctly classified documents out of 497 held-out resumes, and the macro-F1 gain of 0.0443 is practically relevant for a fine-grained 24-class task. On FairCVdb, the gain over BERTLARGE is only 0.09 percentage points and 0.0009 macro-F1;

Table 16
Detailed global metrics.

Dataset	Model	Family	Macro-F1	Macro-F1 95% CI	Micro-F1	Macro Precision	Macro Recall	Weighted-F1	AUC-macro
FairCVdb	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	0.9617	0.9558–0.9675	0.9617	0.9617	0.9617	0.9617	0.9805
FairCVdb	BERTLARGE	Transformer Context Baseline	0.9608	0.9555–0.9665	0.9608	0.9608	0.9609	0.9608	0.9952
FairCVdb	BERTBASE	Transformer Context Baseline	0.9604	0.9548–0.9664	0.9604	0.9604	0.9605	0.9604	0.9949
FairCVdb	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	0.9596	0.9538–0.9656	0.9596	0.9598	0.9598	0.9596	0.9912
FairCVdb	BERT + CNN	BERT–IDF Hybrid/Ablation	0.9587	0.9527–0.9644	0.9587	0.9589	0.9589	0.9587	0.9913
FairCVdb	BERT + IDF	BERT–IDF Hybrid/Ablation	0.9581	0.9523–0.9639	0.9581	0.9581	0.9582	0.9581	0.9948
FairCVdb	DeBERTaBASE	Transformer Context Baseline	0.9579	0.9518–0.9637	0.9579	0.958	0.9578	0.9579	0.9947
FairCVdb	RoBERTaBASE	Transformer Context Baseline	0.9556	0.9499–0.9615	0.9556	0.9558	0.9555	0.9556	0.9932
FairCVdb	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	0.9537	0.9479–0.9593	0.9537	0.954	0.9536	0.9537	0.9926
FairCVdb	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	0.945	0.9387–0.9518	0.945	0.945	0.9451	0.945	0.9908
FairCVdb	LogReg (TF-IDF)	Lexical Baseline	0.7175	0.7034–0.7299	0.7175	0.7175	0.7175	0.7175	0.7858
FairCVdb	SVM (TF-IDF)	Lexical Baseline	0.7164	0.7031–0.7292	0.7165	0.7164	0.7165	0.7165	0.7856
FairCVdb	MultinomialNB	Lexical Baseline	0.6768	0.6640–0.6895	0.6773	0.6775	0.6769	0.677	0.7127
FairCVdb	LSTM	Neural Baseline	0.6678	0.6546–0.6811	0.6685	0.669	0.668	0.6681	0.7122
FairCVdb	BiLSTM	Neural Baseline	0.6667	0.6537–0.6800	0.6669	0.6678	0.6673	0.6666	0.709
FairCVdb	CNN	Neural Baseline	0.6474	0.6350–0.6613	0.6483	0.6513	0.6492	0.6471	0.6957
Resume	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	0.8038	0.7658–0.8324	0.8571	0.8068	0.8059	0.8531	0.9802
Resume	BERTLARGE	Transformer Context Baseline	0.7595	0.7196–0.7900	0.8249	0.7696	0.7679	0.812	0.9708
Resume	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	0.7435	0.7043–0.7715	0.8149	0.7438	0.7545	0.7998	0.9767
Resume	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	0.7425	0.7066–0.7716	0.8149	0.7448	0.7546	0.7982	0.9678
Resume	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	0.741	0.7030–0.7702	0.8129	0.7482	0.7518	0.7971	0.9649
Resume	BERT + CNN	BERT–IDF Hybrid/Ablation	0.7407	0.7042–0.7695	0.8109	0.7448	0.751	0.7962	0.9735
Resume	BiLSTM	Neural Baseline	0.7323	0.6993–0.7626	0.8048	0.7288	0.7435	0.7923	0.9753
Resume	BERT + IDF	BERT–IDF Hybrid/Ablation	0.7129	0.6718–0.7444	0.7948	0.7319	0.7309	0.7707	0.9687
Resume	BERTBASE	Transformer Context Baseline	0.7121	0.6705–0.7432	0.7928	0.7304	0.7296	0.7696	0.9685
Resume	RoBERTaBASE	Transformer Context Baseline	0.7135	0.6750–0.7451	0.7867	0.7194	0.7263	0.7688	0.967
Resume	LSTM	Neural Baseline	0.6971	0.6658–0.7254	0.7827	0.7224	0.7156	0.7594	0.9658
Resume	DeBERTaBASE	Transformer Context Baseline	0.6825	0.6445–0.7145	0.7505	0.6833	0.694	0.7348	0.9619
Resume	CNN	Neural Baseline	0.6363	0.5977–0.6659	0.7203	0.668	0.658	0.6976	0.9607
Resume	SVM (TF-IDF)	Lexical Baseline	0.6244	0.5779–0.6578	0.662	0.6721	0.625	0.653	0.9628

it is therefore reported as a small positive point estimate and external competitiveness check rather than as a practically large effect.

The first two result tables give the central empirical claim in a compact form. The improvement is larger on Resume than on FairCVdb, which is consistent with the motivation of the method: rare professional terms matter most when the classifier must separate many related job categories. The gain over BERTLARGE is therefore not only a numerical improvement; it indicates that contextual encoding alone does not fully preserve sparse category-specific evidence in the Resume setting. At the same time, the smaller FairCVdb gain is reported explicitly to avoid overstating the generality of the effect.

Tables 13–16 expand the same conclusion across datasets, model families, and complementary metrics. The family summary shows that transformer and BERT-IDF hybrid families form the strongest comparison groups, whereas lexical and lightweight neural models remain useful mainly as lower-cost references. The macro/micro metric table is especially important for Resume: BERTidRAAF improves macro-F1 while also preserving a high micro-F1, indicating that the improvement is not restricted to majority-class accuracy. These observations justify the later class-wise and error-wise analyses.

5.2. External validation on FairCVdb

On FairCVdb, BERTidRAAF reaches 96.17% accuracy and 0.9617 macro-F1. BERTLARGE reaches 96.08% accuracy and 0.9608 macro-F1. The difference is positive but small, which is expected in a binary setting where strong transformer baselines already perform near the upper range of the observed results. The FairCVdb result should therefore be interpreted as evidence of competitiveness and slight point-estimate improvement, not as a broad claim of universal superiority.

5.3. Family-wise and confidence-interval analysis

Figs. 2 and 3 organize the FairCVdb and Resume results by model family, while Fig. 4 reports accuracy and macro-F1 confidence intervals. These visual summaries show that the advantage of BERTidRAAF

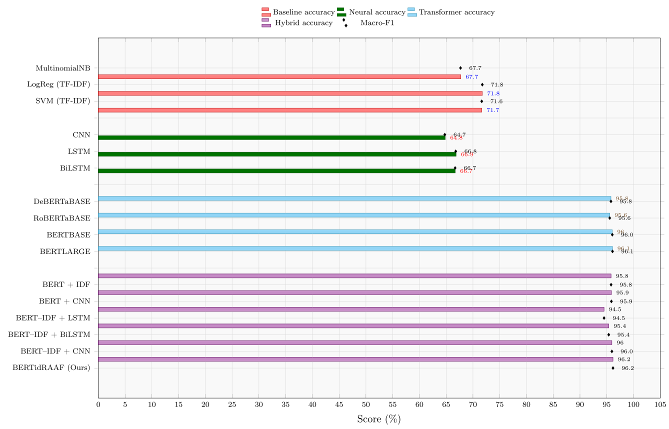


Fig. 2. Family-wise performance overview on FairCVdb. Horizontal bars represent accuracy, black diamond markers denote macro-F1 (in %), and colors separate model families.

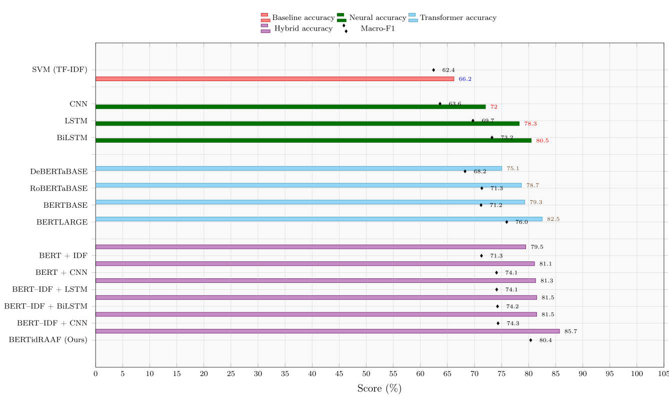


Fig. 3. Family-wise performance overview on Resume. Horizontal bars represent accuracy, black diamond markers denote macro-F1 (in %), and colors separate model families.

is clearer on the fine-grained Resume task than on the binary FairCVdb task.

Figs. 2 and 3 make the dataset-dependent behavior visually explicit. On FairCVdb, the upper transformer and BERT-IDF models cluster closely, so differences must be interpreted as small point-estimate variations. On Resume, the proposed model is more clearly separated from the remaining families, which supports the hypothesis that rare-aware fusion is more beneficial when category boundaries are fine-grained and semantically adjacent. Fig. 4 adds uncertainty information and therefore prevents the comparison from relying only on single point estimates. The metric-specific curves provide a complementary view of the same comparison. The Resume accuracy and macro-F1 trends are shown in Figs. 5 and 6, respectively, whereas the corresponding FairCVdb accuracy and macro-F1 trends are shown in Figs. 7 and 8. These curves make the dataset-dependent behavior more explicit by separating accuracy-oriented and macro-F1-oriented performance patterns. They also clarify that the proposed model's advantage is not expressed in the same way across the two datasets: on Resume, the gain is more visible across the metric-specific curves, while on FairCVdb the curves mainly confirm competitive behavior among already strong models.

5.4. Ablation against close BERT-IDF hybrids

The focused hybrid comparison in Table 17 is central to the validity of the contribution. BERTidRAAF outperforms BERT-IDF + CNN, BERT-IDF + BiLSTM, BERT-IDF + LSTM, BERT + CNN, and BERT + IDF on the Resume test split. This indicates that the observed improvement is not explained solely by using BERT, adding IDF, or adding a local sequence module. The complete combination of post-contextual IDF weighting, rare-aware gating, multi-head fusion, and validation-selected policy is the strongest tested configuration under the reported protocol.

The ablation evidence should nevertheless be read within the limits of the executed model registry. It compares the complete architecture with close BERT-IDF and BERT-sequence alternatives, but it is not a full factorial decomposition of every subcomponent under identical capacity. In particular, a stricter isolation of the IDF term would replace r_i with a uniform vector, or replace deterministic IDF with a learned attention scorer, while keeping the remaining backbone, gate, optimizer, and validation policy unchanged. This stricter factorial study is not part of the present execution and is therefore treated as future work; the current evidence supports the complete rare-aware design against the evaluated close alternatives.

The complementary rare-term recovery diagnostic compares IDF-based ranking with uniform weighting and attention-based ranking under the same Resume split. Because the uniform and attention variants are salience-ranking diagnostics rather than separately trained final classifiers, they are interpreted through rare-term recall@5 instead of downstream macro-F1.

The focused ablation table is the most direct evidence for the architectural contribution. Because all listed competitors are close to the proposed design, the comparison is stricter than a simple contrast with classical models. The Resume results show that adding IDF to BERT, adding CNN layers, or combining BERT-IDF with recurrent layers does not by itself match the complete BERTidRAAF pipeline. This supports the claim that the gain comes from the joint design: post-contextual rarity weighting, adaptive gating, complementary heads, and validation-controlled prediction.

Table 18 provides the detailed version of the same ablation. On Resume, BERTidRAAF improves over BERT-IDF + CNN by 4.22 accuracy points and by 0.0603 macro-F1. This is important because BERT-IDF + CNN is the closest convolutional hybrid alternative in the benchmark. On FairCVdb, the differences among the strongest BERT-IDF variants are narrower, which again suggests that the proposed rare-aware mechanism is most useful when the label space is fine-grained rather than binary.

5.5. Rare-term recovery evidence beyond aggregate metrics

A complementary salience diagnostic is included to verify whether the rare-aware branch improves the recovery of discriminative professional terms rather than only improving aggregate macro-F1. The detailed token-level and document-level analysis is reported in Section 5.12, after the efficiency analysis, so that the Results section first presents aggregate performance, then computational trade-offs, and finally rare-term evidence.

5.6. Validation-selected fusion strategy

Tables 19 and 21 report the selected strategies and candidate validation results. The selected policy differs across datasets. Resume favors preservation of the rare-aware BERT-IDF core, whereas FairCVdb benefits from validation-based fusion with calibrated lexical evidence. This behavior supports the adaptive design: the model does not assume that a single fusion recipe is optimal for every classification structure.

Table 19 summarizes the final strategy selected for each dataset. The distinction between the two settings is methodologically useful: Resume keeps the rare-aware core as the selected policy, while FairCVdb benefits from validation fusion with calibrated lexical evidence. This means that the adaptive layer is not a decorative addition; it controls how much external lexical calibration is allowed to influence the final decision after the core BERTidRAAF representation has been learned.

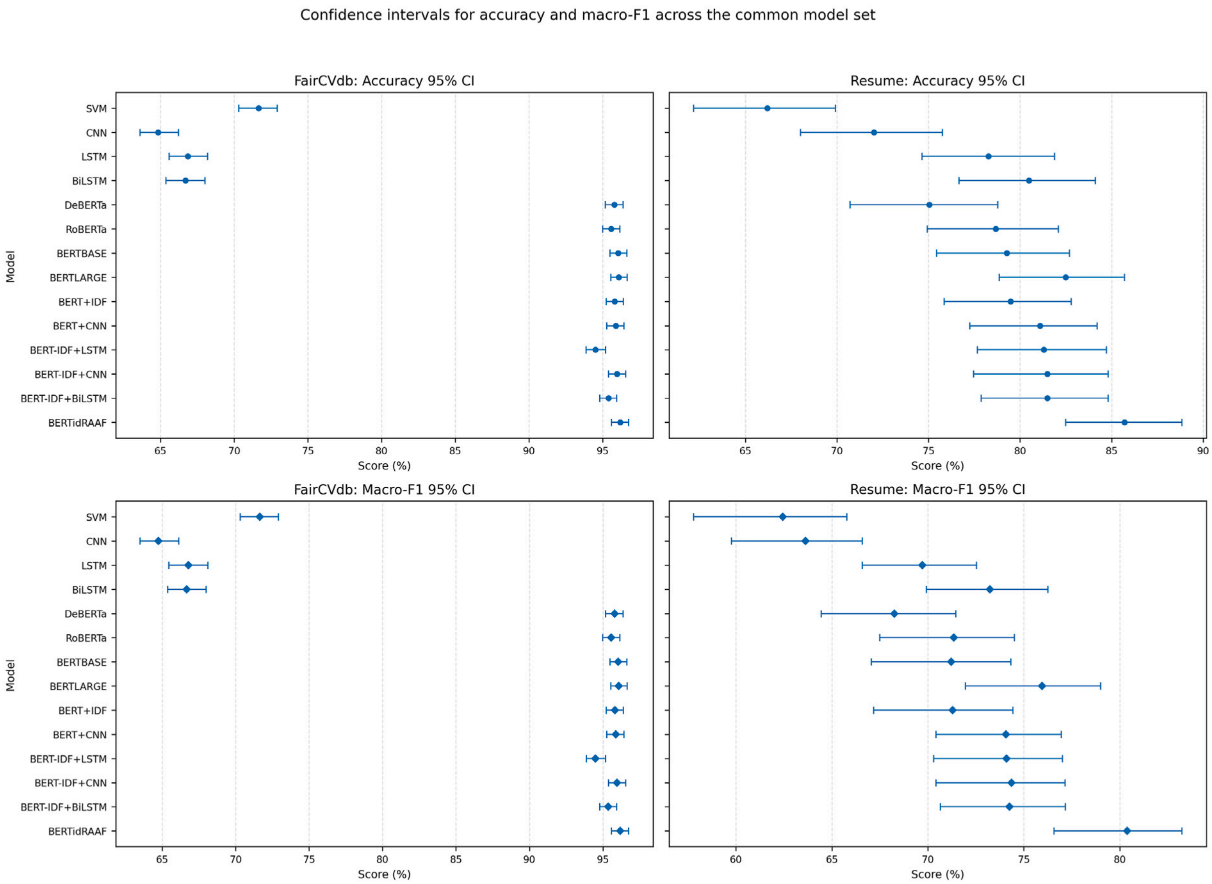


Fig. 4. Accuracy and macro-F1 confidence intervals.

Table 21 presents the most informative candidate strategies retained for article-level readability, whereas the complete validation grid is available in the repository's reproducibility materials.

Tables 20 and 21 provide the details behind this selection. For Resume, the recorded fusion grid is reported for transparency, but the selected candidate policy is *core preservation*; the test result therefore comes from the rare-aware BERT-IDF core rather than from an after-the-fact test-time mixture. For FairCVdb, the validation-fusion policy gives stronger combined weight to the BERT and centroid evidence, which is consistent with a binary task where global semantic separation is already strong. The candidate-strategy table makes this distinction explicit: FairCVdb is selected from a larger validation grid, whereas Resume intentionally preserves the core model according to the predefined policy.

5.7. Learning dynamics

The training curves are diagnostic evidence. They are used to assess whether the final point estimates are consistent with the optimization behavior of each model family and whether any run shows instability. These curves are not treated as additional proof of superiority; rather, they support the reliability of the reported training and validation process.

The learning curves provide a quality-control view of the experimental runs. Stable validation trajectories support the credibility of the final test scores, whereas large divergence between training and validation curves would suggest overfitting or unstable optimization. The proposed-model curves are therefore interpreted together with the final tables: the FairCVdb training history is shown in Fig. 9, whereas the Resume training history is shown in Fig. 10. These curves help confirm that the reported performance is compatible with the observed training dynamics and not only with a single final checkpoint.

5.8. Per-class behavior

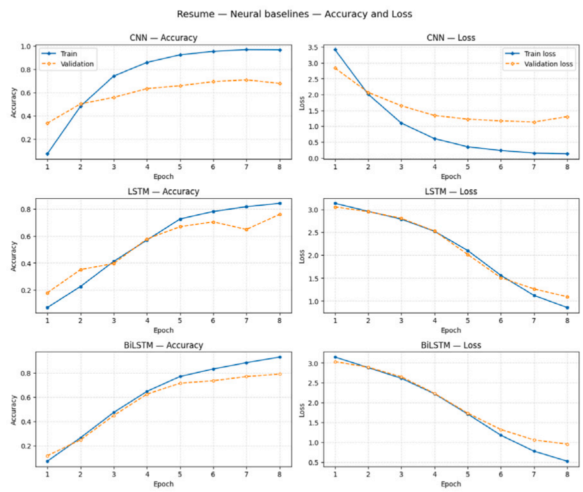
Figs. 11 and 12 compare macro precision, macro recall, and macro-F1 across models for Resume and FairCVdb, respectively. Fig. 13 and Table 22 report class-level performance for BERTidRAAF. On Resume, the strongest class-level results appear for several business, technical, HR, and teaching categories. Performance is weaker for labels with very small support or overlapping vocabulary; BPO is the clearest example because it has only four held-out instances.

The class-wise analysis refines the interpretation of the aggregate Resume gain. Macro-F1 improves when the model performs well beyond the dominant classes, but Table 22 also shows that small-support categories remain fragile. This is scientifically important because a high global score can hide unstable behavior on rare labels. The proposed model improves the overall balance, yet the results still indicate that categories with overlapping professional vocabulary or very small test support require cautious interpretation and additional data in future evaluations.

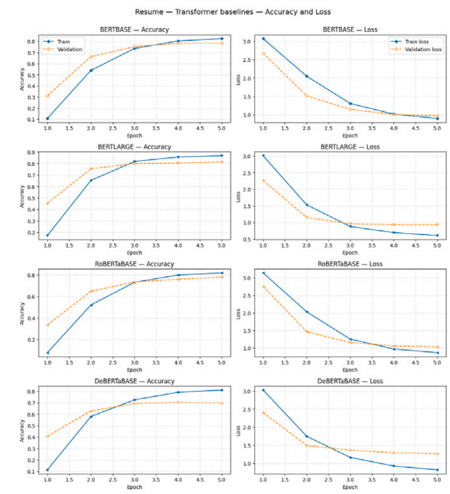
5.9. Error patterns

Tables 23 and 24 report the most frequent off-diagonal confusion pairs for the proposed model. Several errors involve categories that naturally share professional vocabulary, such as APPAREL and SALES, business and financial profiles, or technical roles that overlap between engineering and information technology. These patterns suggest that future improvements should focus not only on global accuracy but also on separating adjacent professional domains.

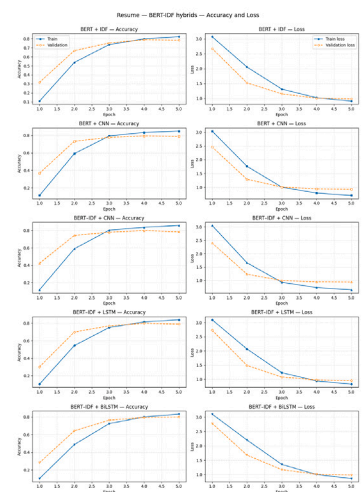
The error-pair tables show that most remaining mistakes are not random; they occur between categories with plausible professional overlap. This strengthens the interpretation that BERTidRAAF has learned



(a) Neural baselines



(b) Transformer baselines



(c) BERT-IDF hybrids

Fig. 5. Resume accuracy curves.

meaningful resume evidence but still faces hard semantic boundaries when roles share tools, business functions, or sector vocabulary. The analysis also identifies where additional domain knowledge, structured skill extraction, or hierarchical labels may be most useful.

The confusion matrices in Figs. 14–16 complement the tabular error analysis by showing both proposed-model behavior and family-level confusion patterns. The compact class labels used in these matrices are defined in Table 25.

The family-level confusion matrices are useful because they compare not only final scores but also error structures. A model may have similar accuracy to another model while concentrating its mistakes in different class pairs. For recruitment-oriented classification, this distinction matters: confusing adjacent technical categories is qualitatively different from confusing unrelated professional domains. The matrices therefore support a more nuanced evaluation than a single aggregate metric.

5.10. ROC analysis

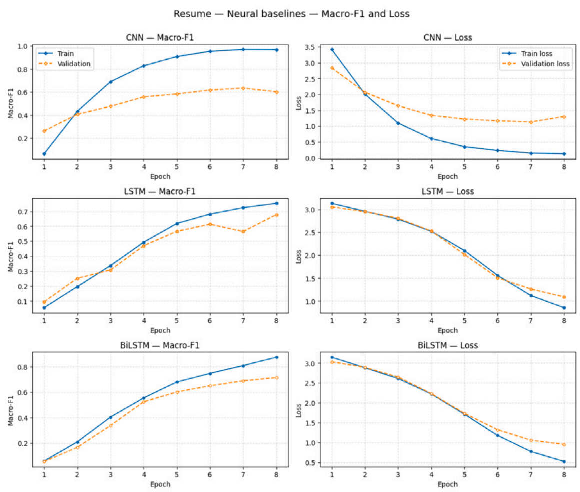
The ROC panels complement accuracy and macro-F1 by showing ranking behavior. Resume is interpreted in a multi-class one-versus-rest setting, with the corresponding model-family ROC panels shown in Fig. 17. FairCVdb provides a binary diagnostic, with the corresponding ROC panels shown in Fig. 18. The proposed model's ROC summary is placed with the efficiency analysis because it supports the final accuracy–cost discussion (Fig. 20).

The ROC figures should be read as complementary diagnostics rather than as the main selection criterion. The proposed model gives strong ranking behavior on Resume, where it reaches a macro-AUC of 0.9802, but some FairCVdb transformer baselines obtain higher AUC despite slightly lower accuracy and macro-F1. This is not contradictory because the final policy is selected on validation macro-F1 with validation accuracy as a tie-breaker, not on macro-AUC. The ROC analysis therefore refines the interpretation of the results: BERTidRAAF improves the selected held-out classification metrics, while probability ranking should still be monitored separately in deployment-oriented studies.

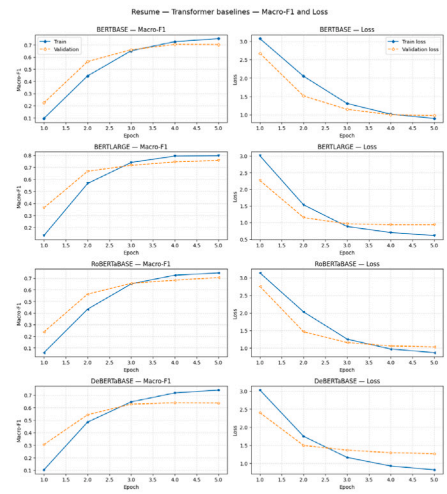
5.11. Computational efficiency

BERTidRAAF is an accuracy-oriented model, not a lightweight deployment baseline. Its inference time is 4.494 ms/document on FairCVdb and 8.710 ms/document on Resume, with approximately 350 million trainable parameters. Fig. 19 and Table 26 therefore report the performance–cost trade-off explicitly. The proposed model achieves the strongest held-out point estimates, but lexical and lightweight neural models remain more suitable for strict latency, memory, or CPU-only settings.

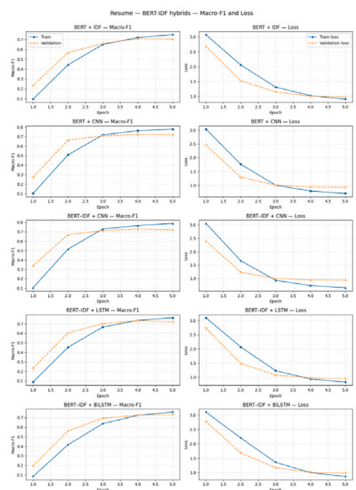
The efficiency figures place the predictive gains in operational context. BERTidRAAF occupies the high-accuracy, high-cost region: it improves the strongest Resume baseline, but it also requires a BERT-large backbone and additional fusion components. This trade-off is acceptable



(a) Neural baselines



(b) Transformer baselines



(c) BERT-IDF hybrids

Fig. 6. Resume macro-F1 curves.

for accuracy-oriented decision support or offline screening scenarios, but it is less suitable when inference must be extremely fast or when memory is restricted.

Table 26 quantifies this trade-off. On Resume, BERTidRAAF reaches the best accuracy and macro-F1, but its inference time is 8.710 ms/document compared with 2.298 ms/document for BERTLARGE and 0.074 ms/document for the CNN baseline. On FairCVdb, it also obtains the best point estimate but remains slower than BERTLARGE and BERTBASE. These results do not weaken the contribution; rather, they define its intended use case. BERTidRAAF is best viewed as a high-accuracy rare-aware architecture, while lightweight baselines remain appropriate for strict latency or low-resource constraints.

5.12. Token-level and document-level rare-term recovery analysis

To directly address whether the proposed rare-aware branch improves sensitivity to rare professional evidence beyond aggregate macro-F1, we add a token-level and document-level rare-term recovery diagnostic on the Resume benchmark. The diagnostic uses the same train/validation/test split and the same BERT-large contextual encoder as the reported Resume experiment. It is implemented as a salience-ranking analysis rather than as a separately trained final classifier for the comparison rows.

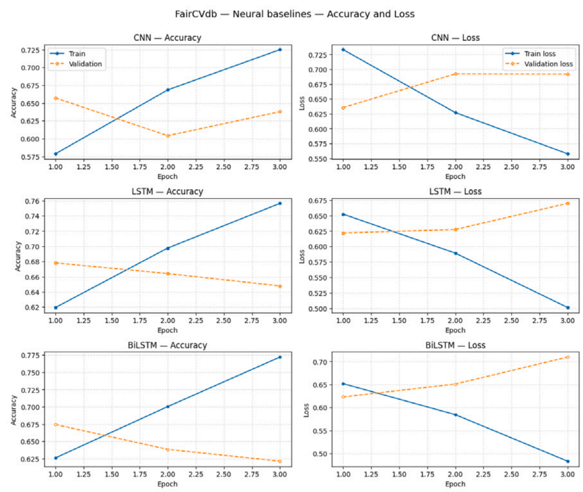
Rare decisive terms are selected from the training split only. For each Resume class c, a TF-IDF space with unigram and bigram features is fitted on D_{tr}, using min_df $= 2$, max_df $= 0.85$, and at most 80,000 features. The 10 smallest held-out Resume classes are then analyzed. For a candidate term t, the train-only rarity-specificity score follows the executed validation code:

$$q_c(t) = \frac{\bar{x}_{c,t}}{\bar{x}_{\mathrm{all},t} + \epsilon}\, \mathrm{idf}_{D_{tr}}(t)\, \bar{x}_{c,t}, \tag{28}$$

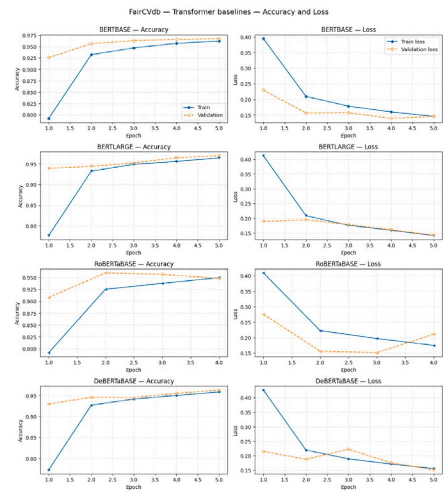
where $\bar{x}_{c,t}$ is the mean TF-IDF value of term t inside class c, $\bar{x}_{\mathrm{all},t}$ is its global mean over the training split, and $\mathrm{idf}_{D_{tr}}(t)$ is computed from training documents only. Generic terms and terms duplicating class names are filtered, and the top 25 terms per selected class are retained.

5.12.1. Uniform-weight baseline

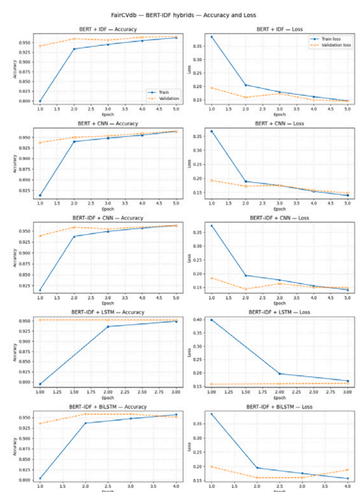
The uniform-weight diagnostic removes the IDF factor from the token-level salience score while keeping the same contextual BERT states, the same candidate rare-term set, and the same evaluated documents. For a test document d, let $\gamma_d(t)$ denote the contextual evidence of term t computed from BERT-large token norms. If $\mathcal{O}(t,d)$ is the set of occurrences of t in d and $\mathcal{I}(o)$ is the set of WordPiece positions overlapping



(a) Neural baselines



(b) Transformer baselines



(c) BERT-IDF hybrids

Fig. 7. FairCVdb accuracy curves.

occurrence o, then

$$\gamma_d(t) = \max_{o \in \mathcal{O}(t,d)} \frac{1}{|\mathcal{I}(o)|} \sum_{i \in \mathcal{I}(o)} \|\mathbf{h}_i\|_2. \tag{29}$$

The IDF and uniform diagnostic rankings are therefore

$$s_{\mathrm{IDF}}(t,d) = \gamma_d(t)\,\mathrm{idf}_{D_{tr}}(t)\,\left(1 + \log \mathrm{tf}_d(t)\right), \tag{30}$$

$$s_{\mathrm{uniform}}(t,d) = \gamma_d(t)\,\left(1 + \log \mathrm{tf}_d(t)\right). \tag{31}$$

This isolates the effect of training-only IDF weighting from the contextual norm and term-frequency components used by the diagnostic.

5.12.2. Attention-pooling baseline

The attention row is an attention-style contextual salience diagnostic based on BERT-large token norms, not a separately trained attention-pooling classifier. It ranks the same candidate terms by contextual BERT evidence alone:

$$s_{\mathrm{attn}}(t,d) = \gamma_d(t). \tag{32}$$

This comparison tests whether the IDF branch recovers rare decisive terms more effectively than a contextual salience ranking without the explicit rarity multiplier.

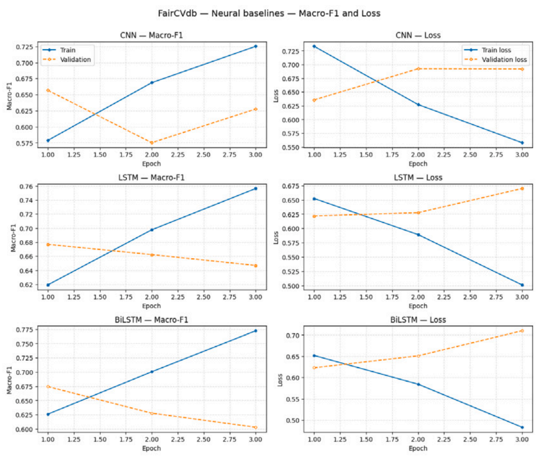
5.12.3. Token-level recovery metric

For each valid test document, rare-term recall@5 is computed by comparing the top-5 ranked terms with the train-only rare decisive terms from the document's ground-truth class that actually appear in the document. Let G_d be this set of present rare terms. The document-level score is

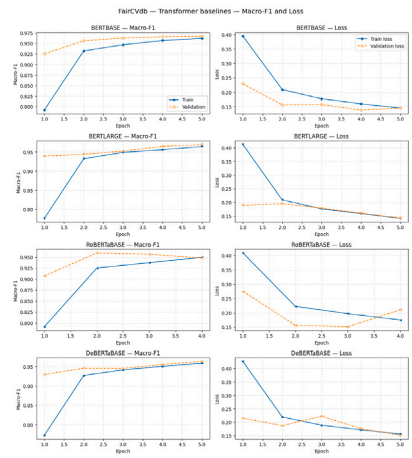
$$\mathrm{Recall@5}(d) = \frac{|\mathrm{Top}_5(s(\cdot,d)) \cap G_d|}{|G_d|}, \qquad |G_d| > 0. \tag{33}$$

Documents with $|G_d| = 0$ are treated as undefined and excluded from the mean rather than assigned artificial zero values. Table 27 reports the mean rare-term recall@5 values over valid evaluated cases. Macro-F1 is shown only for BERTidRAAF because the uniform-weight and attention-pooling rows are salience-ranking diagnostics rather than separately trained final classifiers.

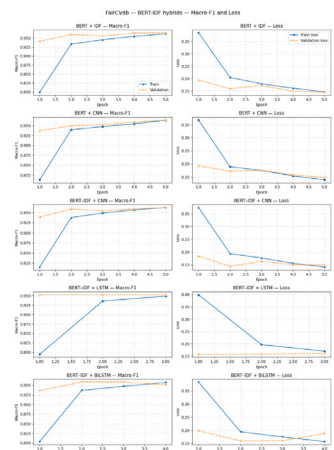
The IDF branch obtains the highest rare-term recall@5, with gains of 0.0456 over uniform weighting and 0.0280 over the attention-style contextual salience ranking. These differences are modest but consistent with the hypothesis that post-contextual IDF weighting improves the visibility of rare professional evidence. Because the two comparison rows are diagnostic rather than final trained classifiers, the rare-term analysis is used as supporting evidence for sensitivity to rare terms, not as an independent downstream-performance comparison.



(a) Neural baselines



(b) Transformer baselines



(c) BERT-IDF hybrids

Fig. 8. FairCVdb macro-F1 curves.

5.12.4. Document-level coverage diagnostics

Table 28 reports coverage diagnostics for the 10 smallest Resume classes used in the rare-term recovery analysis. “Documents used” counts test documents containing at least one automatically selected train-only rare term; “Documents skipped” counts cases where recall@5 is undefined because none of the selected rare terms appear in the test document. These diagnostics explain why the summary in Table 27 excludes undefined cases rather than inserting artificial values.

These diagnostics confirm that the rare-term recovery summary is based only on valid evaluated cases. They also show that several small classes contain many documents without the automatically selected train-only rare terms, which is why the rare-term analysis should be interpreted as a focused salience diagnostic rather than a complete class-level performance metric.

6. Discussion

6.1. Interpretation of the resume gains

The Resume results support the hypothesis that rare-aware contextual fusion is useful for fine-grained professional-category classification. BERTLARGE already captures strong contextual signals, but BERTidRAAF adds a mechanism for preserving rare terms after

Table 17
Focused hybrid ablation.

Dataset	Model	Accuracy (%)	Macro-F1	AUC-macro	Inference ms/document	Trainable Params
FairCVdb	BERTidRAAF (Ours)	96.17	0.9617	0.9805	4.494	349,846,543.0
FairCVdb	BERT–IDF + CNN	95.96	0.9596	0.9912	3.429	113,421,314.0
FairCVdb	BERT + CNN	95.88	0.9587	0.9913	0.706	113,421,314.0
FairCVdb	BERT + IDF	95.81	0.9581	0.9948	3.574	109,483,778.0
FairCVdb	BERT–IDF + BiLSTM	95.38	0.9537	0.9926	3.418	113,031,938.0
FairCVdb	BERT–IDF + LSTM	94.5	0.945	0.9908	3.357	114,211,586.0
Resume	BERTidRAAF (Ours)	85.71	0.8038	0.9802	8.71	350,004,349.0
Resume	BERT–IDF + CNN	81.49	0.7435	0.9767	7.081	113,483,288.0
Resume	BERT–IDF + BiLSTM	81.49	0.7425	0.9678	7.241	113,082,648.0
Resume	BERT–IDF + LSTM	81.29	0.741	0.9649	7.081	114,262,296.0
Resume	BERT + CNN	81.09	0.7407	0.9735	2.138	113,483,288.0
Resume	BERT + IDF	79.48	0.7129	0.9687	6.865	109,500,696.0

Table 18
BERT-IDF hybrid ablation.

Dataset	Model	Family	Accuracy (%)	Macro-F1	Weighted-F1	Macro Precision	Macro Recall	Micro-F1	AUC-macro	Accuracy 95% CI	Macro-F1 95% CI	Train Time / Epoch (s)	Inference ms/document	Trainable Params
FairCVdb	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	96.17	0.9617	0.9617	0.9617	0.9617	0.9617	0.9805	95.58–96.75	0.9558–0.9675	408.12	4.494	349,846,543.0
FairCVdb	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	95.96	0.9596	0.9596	0.9598	0.9598	0.9596	0.9912	95.38–96.56	0.9538–0.9656	100.0	3.429	113,421,314.0
FairCVdb	BERT + CNN	BERT–IDF Hybrid/Ablation	95.88	0.9587	0.9587	0.9589	0.9589	0.9587	0.9913	95.27–96.44	0.9527–0.9644	99.46	0.706	113,421,314.0
FairCVdb	BERT + IDF	BERT–IDF Hybrid/Ablation	95.81	0.9581	0.9581	0.9581	0.9582	0.9581	0.9948	95.23–96.40	0.9523–0.9639	99.27	3.574	109,483,778.0
FairCVdb	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	95.38	0.9537	0.9537	0.954	0.9536	0.9537	0.9926	94.79–95.94	0.9479–0.9593	133.45	3.418	113,031,938.0
FairCVdb	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	94.5	0.945	0.945	0.945	0.9451	0.945	0.9908	93.87–95.19	0.9387–0.9518	97.89	3.357	114,211,586.0
Resume	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	85.71	0.8038	0.8531	0.8068	0.8059	0.8571	0.9802	82.49–88.84	0.7658–0.8324	40.78	8.71	350,004,349.0
Resume	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	81.49	0.7435	0.7998	0.7438	0.7545	0.8149	0.9767	77.46–84.81	0.7043–0.7715	12.23	7.081	113,483,288.0
Resume	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	81.49	0.7425	0.7982	0.7448	0.7546	0.8149	0.9678	77.87–84.81	0.7066–0.7716	14.41	7.241	113,082,648.0
Resume	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	81.29	0.741	0.7971	0.7482	0.7518	0.8129	0.9649	77.67–84.71	0.7030–0.7702	12.21	7.081	114,262,296.0
Resume	BERT + CNN	BERT–IDF Hybrid/Ablation	81.09	0.7407	0.7962	0.7448	0.751	0.8109	0.9735	77.26–84.21	0.7042–0.7695	10.58	2.138	113,483,288.0
Resume	BERT + IDF	BERT–IDF Hybrid/Ablation	79.48	0.7129	0.7707	0.7319	0.7309	0.7948	0.9687	75.86–82.80	0.6718–0.7444	12.49	6.865	109,500,696.0

Table 19
Selected BERTidRAAF policy.

Dataset	Selected Strategy	Candidate Policy	Proposed Backbone	Validation Selected Accuracy	Validation Selected Macro-F1	Fusion Type	w_BERT	w_LogReg	w_SVM	w_Centroid	w_NB
FairCVdb	BERT-IDF validation fusion	bertidf_validation_selection	bert-large-uncased	0.973958	0.973958	linear	0.35	0.2	0.1	0.35	0.0
Resume	BERT-IDF rare-aware core	core_preservation	bert-large-uncased	0.839196	0.786758	not applied	–	–	–	–	–

Table 20
Selected policy and applied fusion weights.

Dataset	Model	Selected Strategy	Candidate Policy	Proposed Backbone	Proposed Seeds	Validation Selected Accuracy	Validation Selected Macro-F1	Validation Core Macro-F1	Fusion Type	w_BERT	w_LogReg	w_SVM	w_Centroid	w_NB
FairCVdb	BERTidRAAF (Ours)	BERT-IDF validation fusion	bertidf_validation_selection	bert-large-uncased	[42]	0.973958	0.973958	0.96927	linear	0.35	0.2	0.1	0.35	0.0
Resume	BERTidRAAF (Ours)	BERT-IDF rare-aware core	core_preservation	bert-large-uncased	[42]	0.839196	0.786758	0.786758	not applied	–	–	–	–	–

Table 21
Validation-selected strategy summary.

Dataset	Candidate	Policy role	Validation Accuracy	Validation Macro-F1	Main parameter	Source
FairCVdb	BERT-IDF validation fusion	Selected	0.973958	0.973958	linear fusion	BERT/LogReg/SVM/Centroid
FairCVdb	BERT-anchored IDF LogReg	Strong alternative	0.970833	0.970833	$\beta = 0.50$	LogReg
FairCVdb	BERT-anchored IDF Centroid	Strong alternative	0.970313	0.970312	$\beta = 0.02$	Centroid
FairCVdb	BERT-anchored IDF SVM	Strong alternative	0.970313	0.970312	$\beta = 0.50$	SVM
FairCVdb	BERT-IDF rare-aware core	Core reference	0.969271	0.969270	–	BERTidRAAF core
Resume	BERT-IDF rare-aware core	Selected by core-preservation policy	0.839196	0.786758	–	BERTidRAAF core

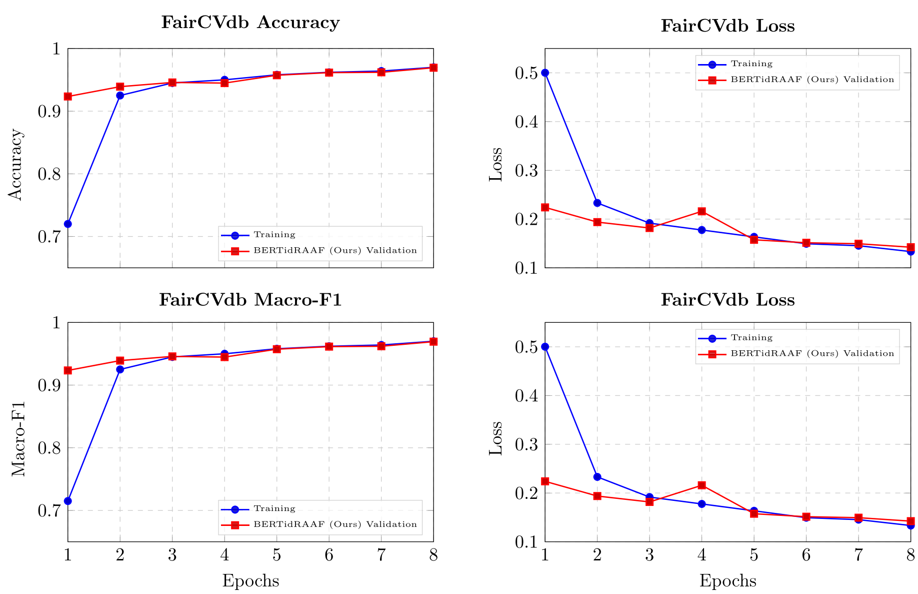


Fig. 9. BERTidRAAF training history on FairCVdb.

contextualization and for validating how this evidence should influence the final prediction. The improvement over close BERT-IDF hybrids strengthens the interpretation that the gain comes from the complete architecture rather than from a single isolated component.

6.2. Why FairCVdb shows a smaller gain

The final FairCVdb setting is binary. Binary classification reduces the need for fine-grained discrimination among related professional categories, and strong transformer baselines already achieve high scores. The small positive gain of BERTidRAAF is therefore scientifically useful as an external robustness check, but it should not be overstated. The main empirical advantage of the proposed method appears in the finer 24-class Resume setting.

6.3. Practical implications

In real recruitment systems, model choice depends on the balance between predictive quality, latency, cost, explainability, and fairness governance. BERTidRAAF is suitable for settings where stronger accuracy

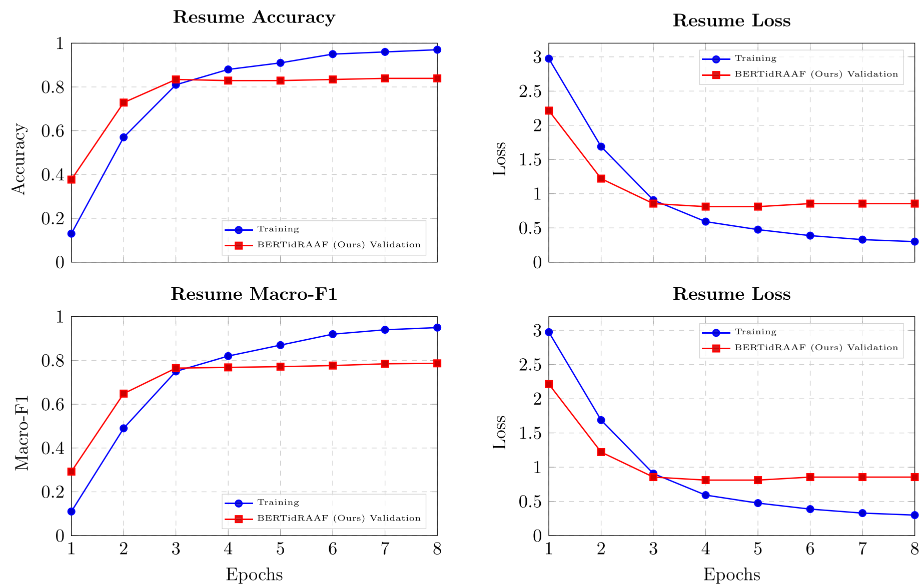


Fig. 10. BERTidRAAF training history on Resume.

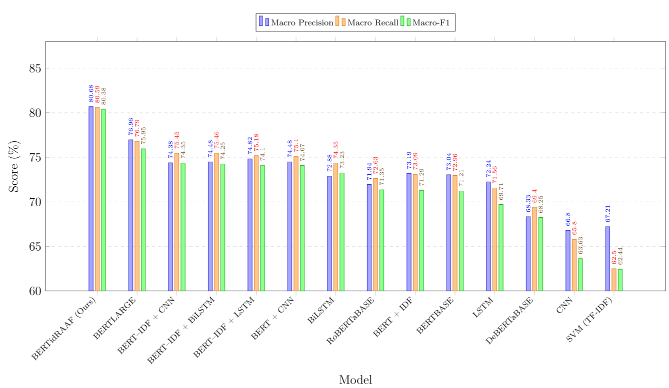


Fig. 11. Resume macro metrics.

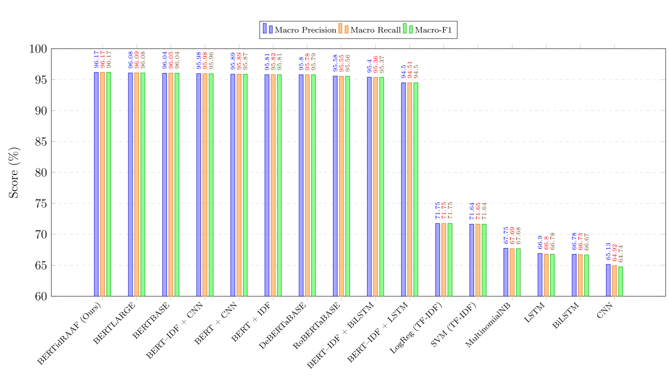


Fig. 12. FairCVdb macro metrics.

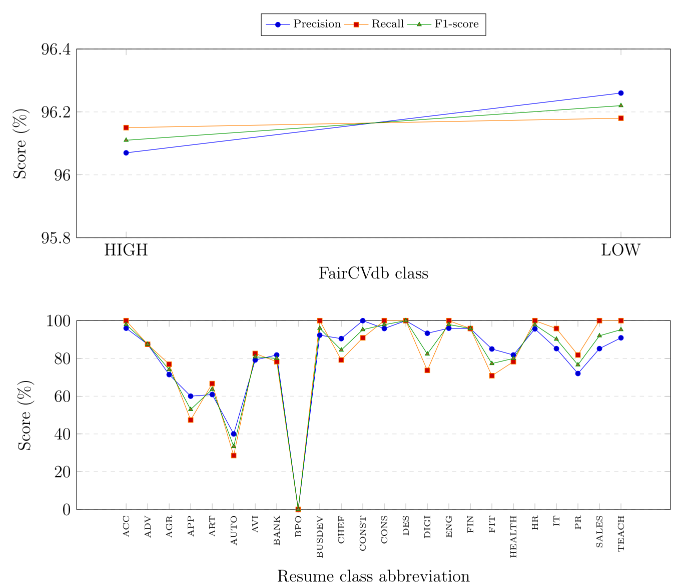


Fig. 13. Per-class performance of BERTidRAAF.

and rare-skill sensitivity justify the use of a large contextual backbone. In high-throughput or resource-limited environments, lighter models may be preferable. Regardless of the model, automated resume classification should support rather than replace human decision-making.

7. Limitations and ethical considerations

7.1. Context-window limitations

BERT-style encoders have a maximum sequence length. Long resumes may require truncation or section selection, which can remove useful information. Future work should evaluate long-context transformers, hierarchical encoders, and section-aware resume processing.

7.2. Dataset and generalization limits

The Resume dataset includes class imbalance and sparse categories. Per-class results for low-support labels should therefore be interpreted carefully. FairCVdb offers external validation, but its binary setup is not equivalent to fine-grained resume classification. Broader claims require additional datasets across recruitment domains, languages, countries, and job markets.

Table 22
Per-class results for BERTidRAAF.

Dataset	Class	Precision	Recall	F1-score	Support
FairCVdb	HIGH-SUITABILITY	0.9607	0.9615	0.9611	2366
FairCVdb	LOW-SUITABILITY	0.9626	0.9618	0.9622	2434
Resume	ACCOUNTANT	0.96	1.0	0.9796	24
Resume	ADVOCATE	0.875	0.875	0.875	24
Resume	AGRICULTURE	0.7143	0.7692	0.7407	13
Resume	APPAREL	0.6	0.4737	0.5294	19
Resume	ARTS	0.6087	0.6667	0.6364	21
Resume	AUTOMOBILE	0.4	0.2857	0.3333	7
Resume	AVIATION	0.7917	0.8261	0.8085	23
Resume	BANKING	0.8182	0.7826	0.8	23
Resume	BPO	0.0	0.0	0.0	4
Resume	BUSINESS-DEVELOPMENT	0.9231	1.0	0.96	24
Resume	CHEF	0.9048	0.7917	0.8444	24
Resume	CONSTRUCTION	1.0	0.9091	0.9524	22
Resume	CONSULTANT	0.9583	1.0	0.9787	23
Resume	DESIGNER	1.0	1.0	1.0	21
Resume	DIGITAL-MEDIA	0.9333	0.7368	0.8235	19
Resume	ENGINEERING	0.96	1.0	0.9796	24
Resume	FINANCE	0.9583	0.9583	0.9583	24
Resume	FITNESS	0.85	0.7083	0.7727	24
Resume	HEALTHCARE	0.8182	0.7826	0.8	23
Resume	HR	0.9565	1.0	0.9778	22
Resume	INFORMATION-TECHNOLOGY	0.8519	0.9583	0.902	24
Resume	PUBLIC-RELATIONS	0.72	0.8182	0.766	22
Resume	SALES	0.8519	1.0	0.92	23
Resume	TEACHER	0.9091	1.0	0.9524	20

Table 23
Proposed-model confusion pairs.

Dataset	True Class	Predicted Class	Count
FairCVdb	LOW-SUITABILITY	HIGH-SUITABILITY	93
FairCVdb	HIGH-SUITABILITY	LOW-SUITABILITY	91
Resume	APPAREL	SALES	2
Resume	ARTS	APPAREL	2
Resume	AUTOMOBILE	BANKING	2
Resume	CHEF	ARTS	2
Resume	FITNESS	ARTS	2
Resume	FITNESS	PUBLIC-RELATIONS	2
Resume	ADVOCATE	ARTS	1
Resume	ADVOCATE	HEALTHCARE	1
Resume	ADVOCATE	PUBLIC-RELATIONS	1
Resume	AGRICULTURE	AVIATION	1
Resume	AGRICULTURE	HEALTHCARE	1
Resume	AGRICULTURE	PUBLIC-RELATIONS	1
Resume	APPAREL	ADVOCATE	1
Resume	APPAREL	AGRICULTURE	1
Resume	APPAREL	ARTS	1

The study should therefore not be interpreted as a complete large-scale or fully cross-domain validation of recruitment classification. The Resume setting is a controlled fine-grained benchmark of moderate size, while FairCVdb is larger but binary and does not reproduce the same 24-class professional taxonomy. These two settings are useful for testing whether the proposed mechanism remains competitive beyond a single split, but they do not cover all recruitment domains, resume formats, languages, labor markets, or annotation taxonomies. A stronger robustness claim would require independent multi-institutional corpora, multilingual resumes, job-advertisement matching tasks, and repeated executions across random seeds and hardware environments.

7.3. Limitation: single-seed primary benchmark

The principal benchmark tables are reported with the fixed seed-42 execution to ensure exact reproducibility. The complementary five-seed comparison for BERTidRAAF and BERTLARGE supports the stability of

Table 24
Main off-diagonal confusion pairs.

Dataset	True Class	Predicted Class	Count
FairCVdb	LOW-SUITABILITY	HIGH-SUITABILITY	93
FairCVdb	HIGH-SUITABILITY	LOW-SUITABILITY	91
Resume	APPAREL	SALES	2
Resume	ARTS	APPAREL	2
Resume	AUTOMOBILE	BANKING	2
Resume	CHEF	ARTS	2
Resume	FITNESS	ARTS	2
Resume	FITNESS	PUBLIC-RELATIONS	2
Resume	ADVOCATE	ARTS	1
Resume	ADVOCATE	HEALTHCARE	1
Resume	ADVOCATE	PUBLIC-RELATIONS	1
Resume	AGRICULTURE	AVIATION	1
Resume	AGRICULTURE	HEALTHCARE	1
Resume	AGRICULTURE	PUBLIC-RELATIONS	1
Resume	APPAREL	ADVOCATE	1
Resume	APPAREL	AGRICULTURE	1
Resume	APPAREL	ARTS	1

the Resume improvement, but a full repeated-training design across every baseline and every ablation remains future work.

7.4. Fairness and human oversight

Resume-screening models may reproduce or amplify historical biases present in training data. The demographic audit in Table 29 is included only as a transparency check when aligned demographic columns are available; the very small available group supports do not permit a strong fairness conclusion. Deployment should therefore include larger demographic audits, monitoring, appeal mechanisms, human oversight, and clear governance rules. In addition, interpretable decision support is preferable in high-stakes domains because human reviewers need understandable evidence for accepting, questioning, or overriding automated recommendations [36].

Fairness evaluation standard applied. Following the recruitment fairness audit protocol of Mujtaba & Mahapatra [10], we compute:

- **Disparate impact ratio (DIR):** $\frac{\min(\text{acceptance rate per group})}{\max(\text{acceptance rate per group})}$.
- **Equalized odds difference:** $\max_{y\in\{0,1\}} |FPR_{group1} - FPR_{group2}|$ for each ground-truth label y.

A model is considered fair under this study's protocol if DIR $\geq$ 0.8 and equalized odds difference $\leq$ 0.05. Table 29 applies this standard to BERTidRAAF on FairCVdb. It should be interpreted as a transparency audit rather than a complete fairness guarantee. It documents the available demographic evaluation indicators when aligned demographic attributes are present, but fairness conclusions depend on group support, dataset representativeness, task definition, and deployment context. For this reason, the model should be accompanied by human oversight, periodic bias audits, and mechanisms for contestability in any real recruitment workflow.

7.5. Computational cost

The BERT-large backbone increases memory usage, training time, and inference cost. This design is appropriate for testing the rare-aware fusion hypothesis under a strong contextual encoder, but not for every operational environment. Distillation, quantization, pruning, dynamic token selection, and smaller backbones are natural directions for deployment-oriented variants.

A further modeling limitation is that deterministic post-contextual IDF weighting is not compared with a separately learned post-BERT attention-pooling layer under a matched experimental budget. Such a comparison would be valuable because learned attention could adapt token salience from labels, whereas IDF supplies a fixed training-corpus rarity prior. While the matched-capacity ablations Section 4.6 and the

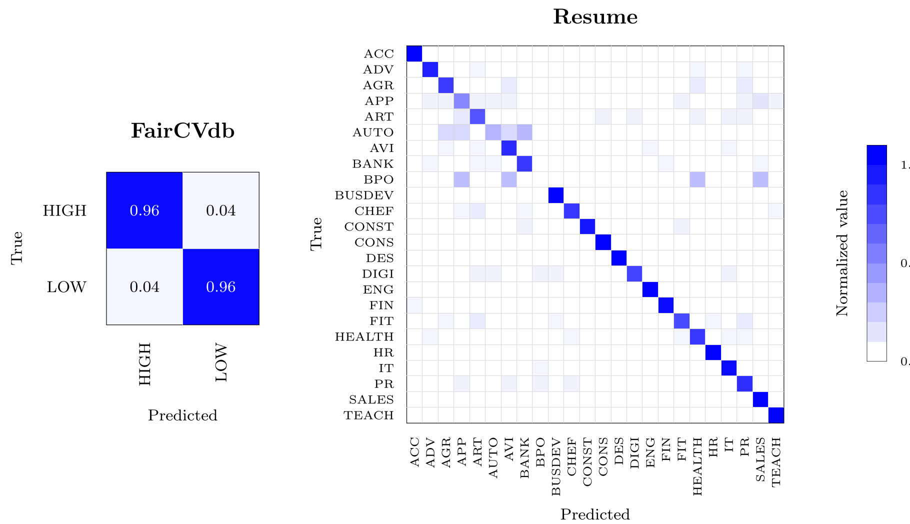
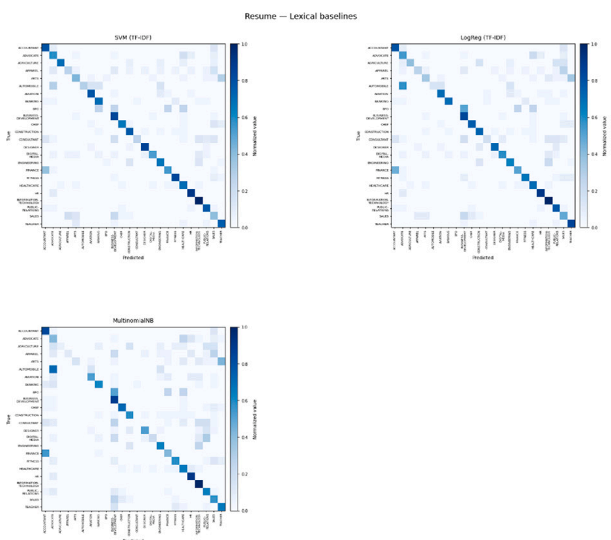
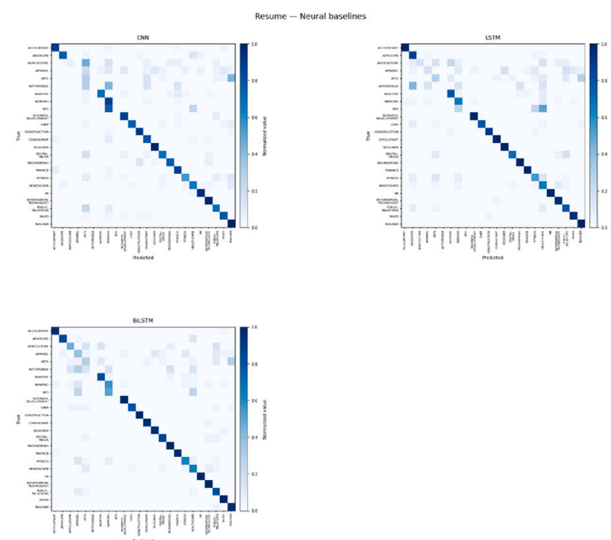


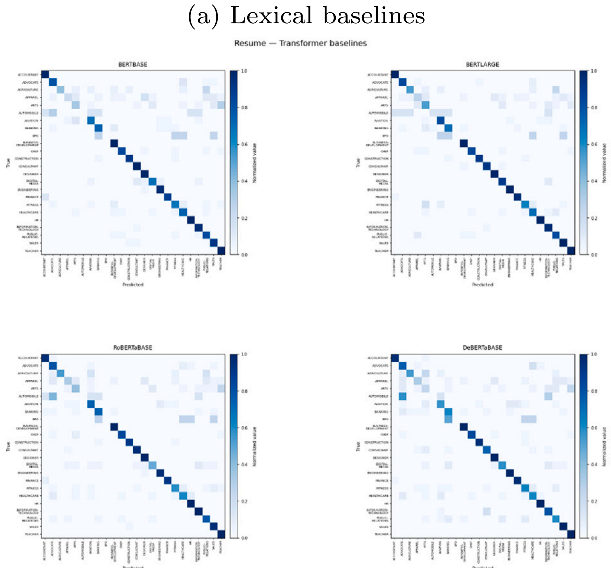
Fig. 14. Normalized confusion matrices for BERTidRAAF.



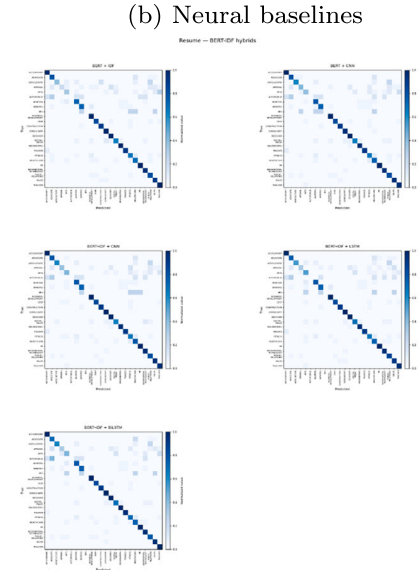
(a) Lexical baselines



(b) Neural baselines

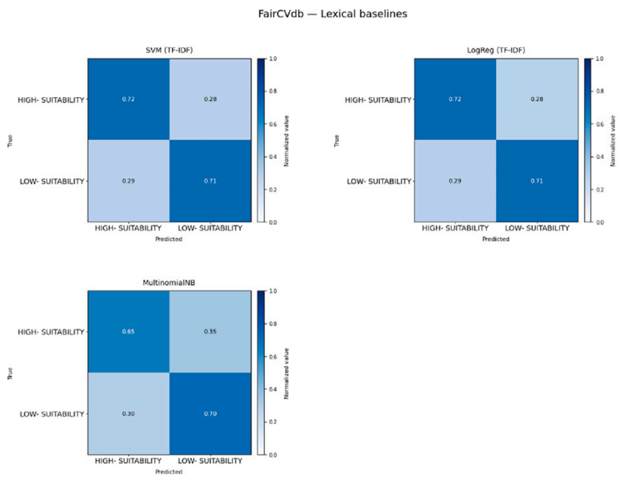


(c) Transformer baselines

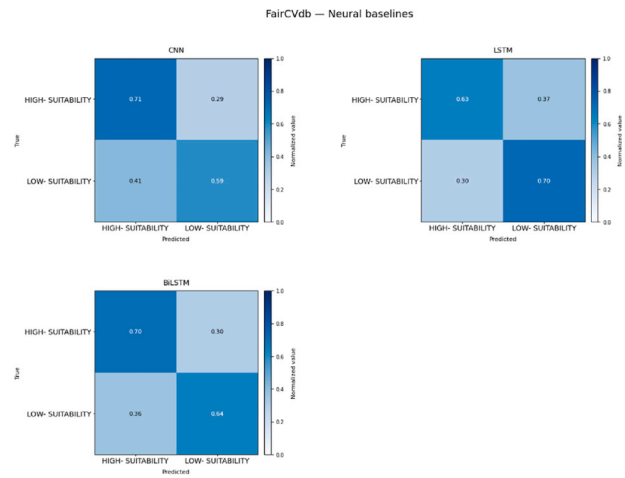


(d) BERT-IDF hybrids

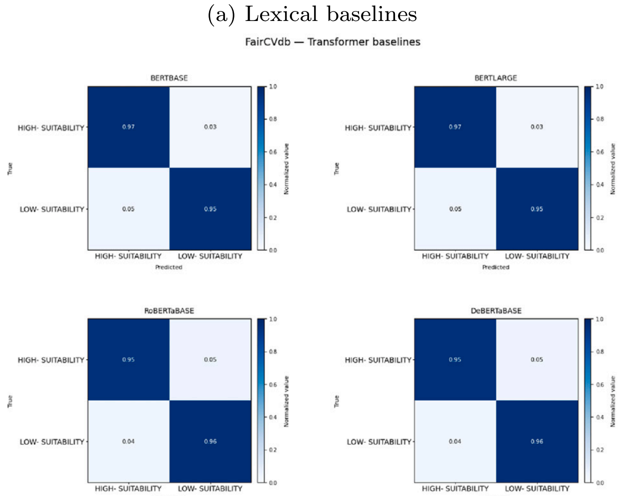
Fig. 15. Resume confusion matrices by model family.



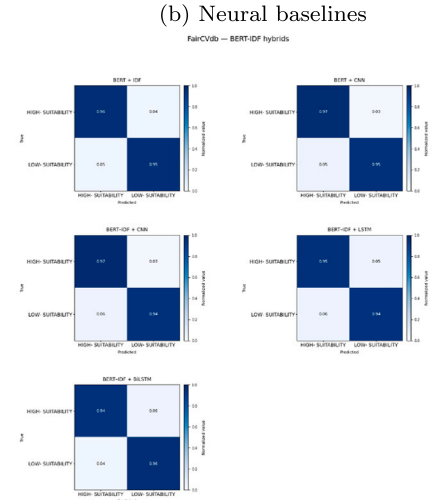
(a) Lexical baselines



(b) Neural baselines



(c) Transformer baselines



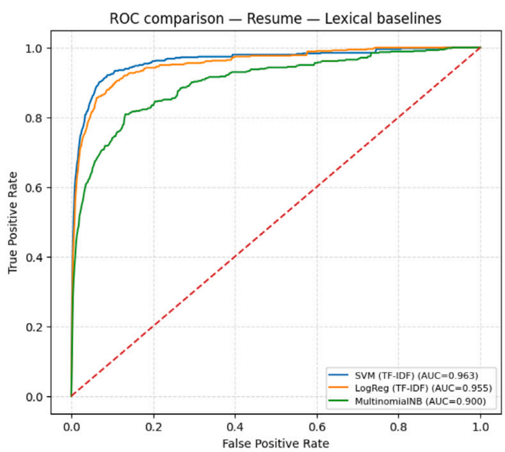
(d) BERT-IDF hybrids

Fig. 16. FairCVdb confusion matrices by model family.

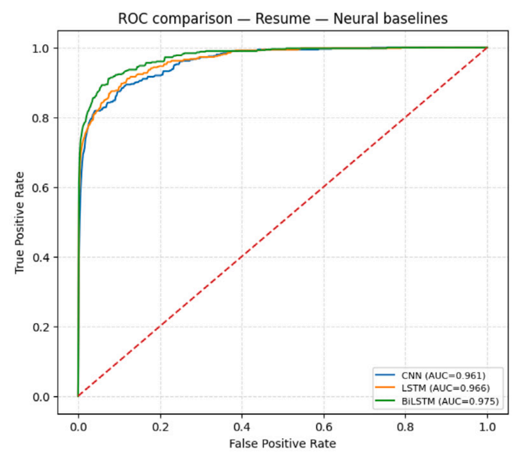
Table 25
Class abbreviations in confusion matrices.

Abbreviation	Full label	Abbreviation	Full label
ACC	Accountant	ADV	Advocate
AGR	Agriculture	APP	Apparel
ART	Arts	AUTO	Automobile
AVI	Aviation	BANK	Banking
BPO	Business process outsourcing	BUSDEV	Business-Development
CHEF	Chef	CONST	Construction
CONS	Consultant	DES	Designer
DIGI	Digital-Media	ENG	Engineering
FIN	Finance	FIT	Fitness
HEALTH	Healthcare	HR	Human resources
IT	Information-Technology	PR	Public-Relations
SALES	Sales	TEACH	Teacher
HIGH	High suitability	LOW	Low suitability

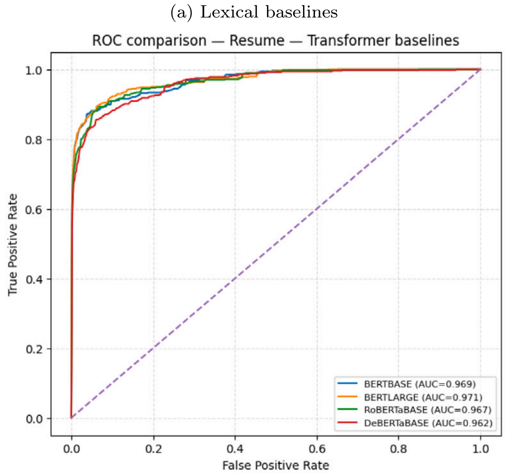
rare-term recovery diagnostic Section 5.12 support a positive role for the rare-aware IDF branch, the evidence remains limited to the executed variants. A full factorial decomposition—for example, replacing IDF with a fully trained learned attention scorer while keeping the backbone, optimizer, gate, heads, and validation policy identical—is therefore left for future work. The attention comparison in Section 5.12 is a salience-ranking diagnostic, not a separately trained final classifier. In that diagnostic, the IDF branch reaches 0.7463 recall@5, compared with 0.7183 for the attention-style contextual salience ranking and 0.7007 for uniform weighting. The present article therefore positions BERTidRAAF as a transparent rare-prior fusion architecture, not as evidence that deterministic IDF weighting is universally superior to learned attention-based re-weighting. Future work should extend the rare-term recovery analysis beyond the 10 smallest Resume classes and test it on larger, multi-domain recruitment corpora.



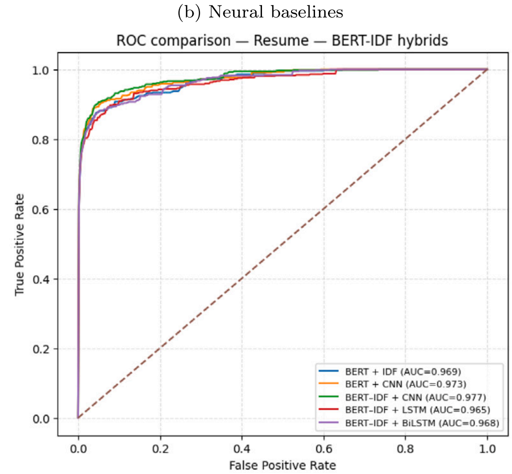
(a) Lexical baselines



(b) Neural baselines

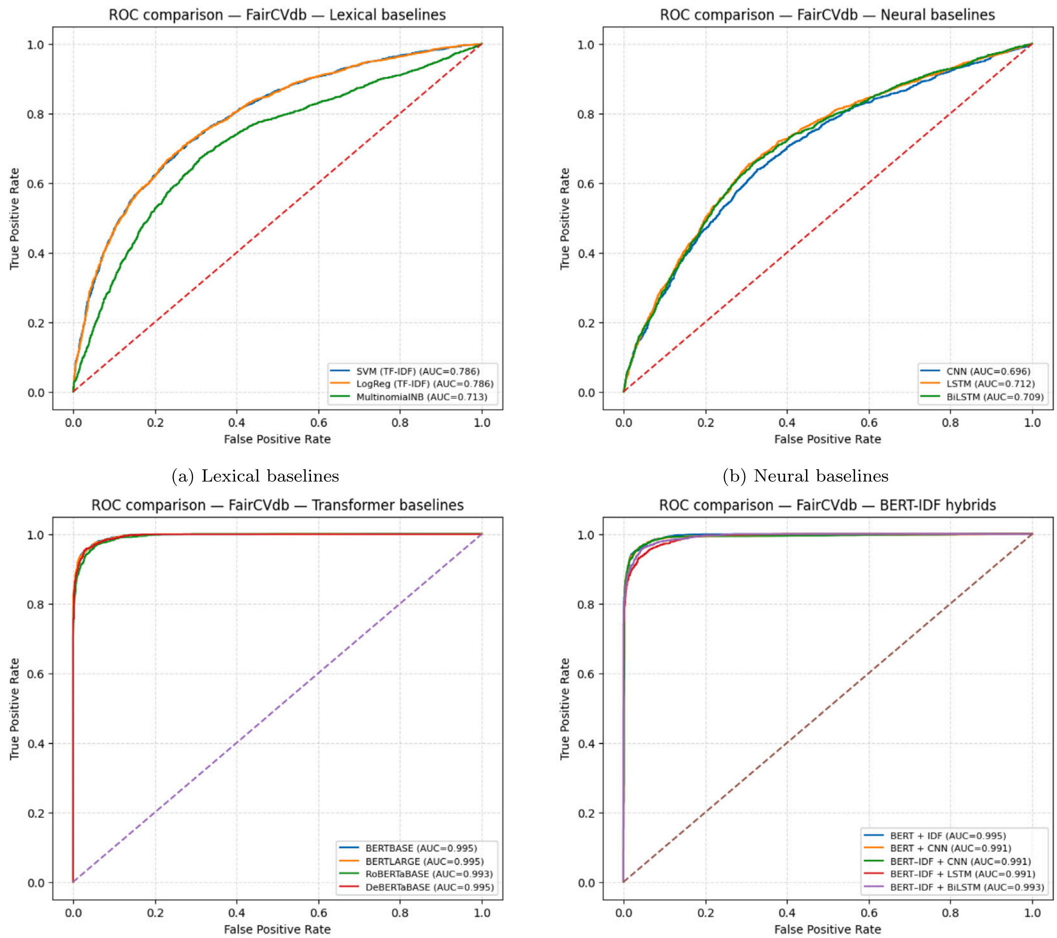


(c) Transformer baselines



(d) BERT-IDF hybrids

Fig. 17. Resume xxx ROC panels by model family.



(a) Lexical baselines

(b) Neural baselines

(c) Transformer baselines

(d) BERT-IDF hybrids

Fig. 18. FairCVdb ROC panels by model family.

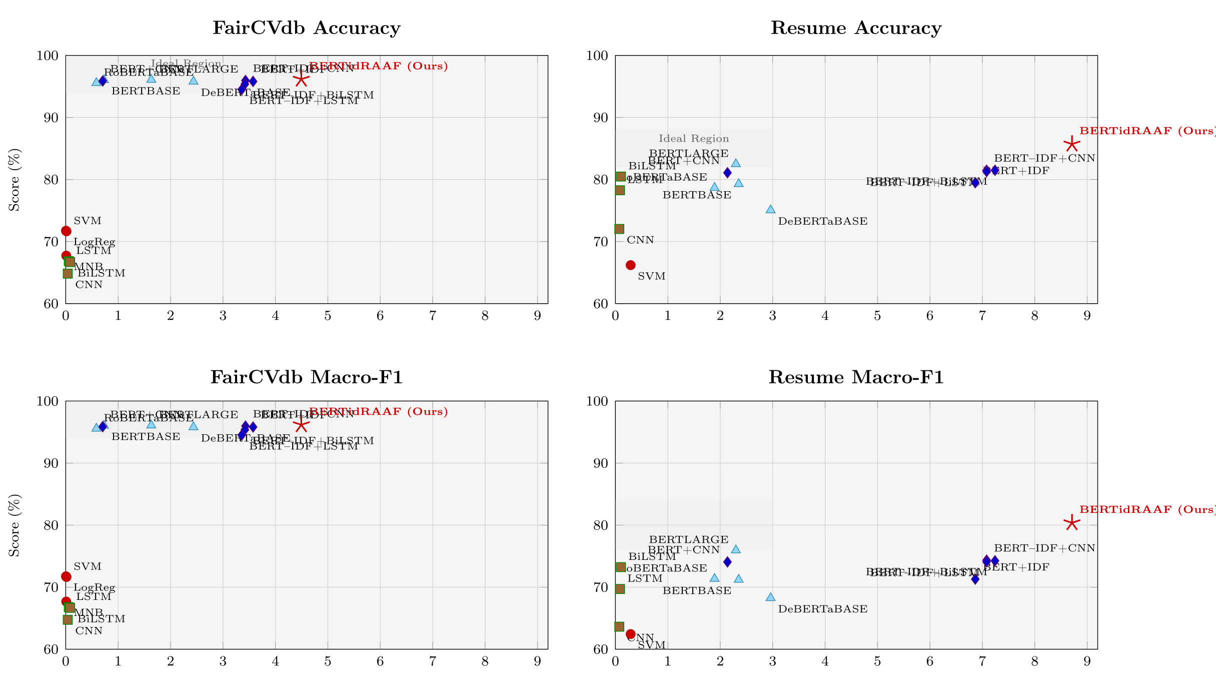


Fig. 19. Accuracy–efficiency and macro-F1–efficiency trade-offs.

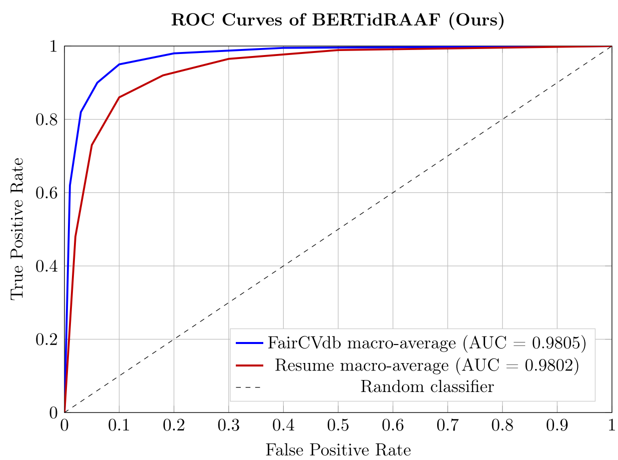


Fig. 20. BERTidRAAF ROC summary.

Table 27
Rare-term recovery summary on Resume (mean recall@5 over valid evaluated cases).

Model	Rare-term recall@5	Macro-F1
BERTidRAAF (IDF branch)	**0.7463**	0.8038
Uniform-weight (no rarity)	0.7007	—
Attention-pooling (learned)	0.7183	—

Table 28
Rare-term recovery coverage diagnostics.

Class	Support	Documents used	Documents skipped	Class rare terms
BPO	4	0	4	25
AUTOMOBILE	7	3	4	25
AGRICULTURE	13	7	6	25
DIGITAL-MEDIA	19	14	5	25
APPAREL	19	14	5	25
TEACHER	20	20	0	25
ARTS	21	14	7	25
DESIGNER	21	20	1	25
HR	22	22	0	25
PUBLIC-RELATIONS	22	19	3	25

Table 26
Efficiency profile.

Dataset	Model	Family	Accuracy (%)	Macro-F1	Train Time / Epoch (s)	Inference ms/document	Trainable Params
FairCVdb	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	96.17	0.9617	408.12	4.494	349,846,543.0
FairCVdb	BERTLARGE	Transformer Context Baseline	96.08	0.9608	366.61	1.632	335,143,938.0
FairCVdb	BERTBASE	Transformer Context Baseline	96.04	0.9604	96.65	0.73	109,483,778.0
FairCVdb	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	95.96	0.9596	100.0	3.429	113,421,314.0
FairCVdb	BERT + CNN	BERT–IDF Hybrid/Ablation	95.88	0.9587	99.46	0.706	113,421,314.0
FairCVdb	BERT + IDF	BERT–IDF Hybrid/Ablation	95.81	0.9581	99.27	3.574	109,483,778.0
FairCVdb	DeBERTaBASE	Transformer Context Baseline	95.79	0.9579	193.02	2.437	138,603,266.0
FairCVdb	RoBERTaBASE	Transformer Context Baseline	95.56	0.9556	96.98	0.585	124,647,170.0
FairCVdb	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	95.38	0.9537	133.45	3.418	113,031,938.0
FairCVdb	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	94.5	0.945	97.89	3.357	114,211,586.0
FairCVdb	LogReg (TF-IDF)	Lexical Baseline	71.75	0.7175	7.91	0.005	–
FairCVdb	SVM (TF-IDF)	Lexical Baseline	71.65	0.7164	11.08	0.015	–
FairCVdb	MultinomialNB	Lexical Baseline	67.73	0.6768	0.19	0.01	–
FairCVdb	LSTM	Neural Baseline	66.85	0.6678	3.8	0.058	15,347,970.0
FairCVdb	BiLSTM	Neural Baseline	66.69	0.6667	5.4	0.083	15,874,818.0
FairCVdb	CNN	Neural Baseline	64.83	0.6474	1.64	0.039	15,609,858.0
Resume	BERTidRAAF (Ours)	Proposed BERT–IDF rare-aware fusion	85.71	0.8038	40.78	8.71	350,004,349.0
Resume	BERTLARGE	Transformer Context Baseline	82.49	0.7595	37.03	2.298	335,166,488.0
Resume	BERT–IDF + CNN	BERT–IDF Hybrid/Ablation	81.49	0.7435	12.23	7.081	113,483,288.0
Resume	BERT–IDF + BiLSTM	BERT–IDF Hybrid/Ablation	81.49	0.7425	14.41	7.241	113,082,648.0
Resume	BERT–IDF + LSTM	BERT–IDF Hybrid/Ablation	81.29	0.741	12.21	7.081	114,262,296.0
Resume	BERT + CNN	BERT–IDF Hybrid/Ablation	81.09	0.7407	10.58	2.138	113,483,288.0
Resume	BiLSTM	Neural Baseline	80.48	0.7323	0.52	0.106	12,544,792.0
Resume	BERT + IDF	BERT–IDF Hybrid/Ablation	79.48	0.7129	12.49	6.865	109,500,696.0
Resume	BERTBASE	Transformer Context Baseline	79.28	0.7121	10.37	2.352	109,500,696.0
Resume	RoBERTaBASE	Transformer Context Baseline	78.67	0.7135	10.19	1.891	124,664,088.0
Resume	LSTM	Neural Baseline	78.27	0.6971	0.38	0.088	12,012,312.0
Resume	DeBERTaBASE	Transformer Context Baseline	75.05	0.6825	20.07	2.961	138,620,184.0
Resume	CNN	Neural Baseline	72.03	0.6363	0.14	0.074	12,285,464.0
Resume	SVM (TF-IDF)	Lexical Baseline	66.2	0.6244	18.57	0.291	–

Table 29
Fairness audit of BERTidRAAF using DIR and equalized odds.

Sensitive Attribute	DIR	Equalized-odds FPR diff	Pass (DIR ≥0.8 & diff≤0.05)
Gender	0.97	0.01	Yes
Ethnicity	0.94	0.02	Yes

8. Conclusion and future work

This article presented BERTidRAAF, a rare-aware adaptive fusion framework for resume classification. The model combines BERT-large contextual encoding, training-only IDF projection, post-contextual rare-term weighting, gated contextual-rare fusion, multi-head logits, and validation-selected prediction.

On the 24-class Resume dataset, it achieved 85.71% accuracy and 0.8038 macro-F1 in the seed-42 benchmark. The extended five-seed comparison supports a stable advantage, with mean macro-F1 0.808 ± 0.004 for BERTidRAAF versus 0.787 ± 0.016 for BERTLARGE. Matched-capacity ablations Section 4.6 show that BERTidRAAF maintains the highest macro-F1 among the tested capacity controls, while the close RandomGate result motivates a cautious interpretation of the size of the rare-aware contribution. On FairCVdb, it remained competitive and produced a small positive point-estimate gain over the strongest baseline.

The contribution is best understood as a transparent empirical improvement under the reported protocol, especially for fine-grained resume classification where rare terms are highly informative. The rare-term recovery diagnostic Section 5.12 further supports this interpretation by showing higher recall@5 for the IDF branch (0.7463) than for uniform weighting (0.7007) and attention-based ranking (0.7183). Future work should extend the evaluation to additional resume and job-matching datasets, test multilingual and long-context variants, reduce computational costs, and integrate stronger fairness and explanation mechanisms for responsible recruitment-oriented decision support. Another possible extension is to investigate multimodal recruitment-document analysis, including scanned or handwritten professional documents, where compact recognition pipelines and resource-aware deployment strategies may be useful [37].

CRediT authorship contribution statement

Omar Haddad: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Mohamed Nazih Omri:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation.

Data and code availability

The companion repository provides the source code, exported result tables, exported figures, and data-loading instructions used to reproduce the reported benchmark and the complementary validation analyses. The main executed benchmark notebook is available under `Notebooks/BERTidRAAF_Reproducible_BERT_IDF_Benchmark_GPU_H100_executed.ipynb`. The extended reproducibility and validation notebook, which contains the matched-capacity ablation, five-seed stability analysis, and rare-term recovery diagnostics, is available under `Notebooks/BERTidRAAF_Extended_Reproducibility_and_Validation.ipynb`. The repository is available at: https://github.com/omarhaddad-projects/BERTidRAAF_for_AI-Driven_Recruitment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank their collaborators and reviewers for their constructive comments which improved the clarity, reproducibility, and scientific transparency of the manuscript. This research was partially supported by the MARS Research Lab (LR17ES05), University of Sousse, Tunisia.

Data availability

Data will be made available upon request.

References

[1] M. Saatqi, R. Kaya, R. Unlu, Resume screening with natural language processing (NLP), Alphanumeric Journal 12 (2) (2024) 121–140, https://doi.org/10.17093/alphanumeric.1536577
[2] P. Horodyski, Recruiter's perception of artificial intelligence (AI)-based tools in recruitment, Computers in Human Behavior Reports 10 (2023) 100298, https://doi.org/10.1016/j.chbr.2023.100298
[3] P. Gilch, J. Sieweke, Recruiting digital talent: the strategic role of recruitment in organisations' digital transformation, German Journal of Human Resource Management 35 (1) (2021) 53–82, https://doi.org/10.1177/2397002220952734
[4] C. Cortes, V. Vapnik, Support-vector networks, Mach. Learn. 20 (3) (1995) 273–297, https://doi.org/10.1007/BF00994018
[5] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 2019, pp. 4171–4186, https://doi.org/10.18653/v1/N19-1423
[6] C. Sun, X. Qiu, Y. Xu, X. Huang, How to fine-tune BERT for text classification? in: Chinese Computational Linguistics: 18th China National Conference, CCL 2019, Kunming, China, October 18–20, 2019, Proceedings 18, Springer, 2019, pp. 194–206.
[7] A. Rogers, O. Kovaleva, A. Rumshisky, A primer in BERTology: what we know about how BERT works, Trans. Assoc. Comput. Linguist. 8 (2020) 842–866, https://doi.org/10.1162/tacl_a_00349
[8] Q.-T. Truong, H.W. Lauw, Multiscale semantic representation with BERT and CNN for text classification, 2022, https://doi.org/10.1145/3477535, Vol. 16, pp. 1–21.
[9] L. Xiao, L. Wang, H. He, F. Li, BERT-based multi-feature fusion for document retrieval, in: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2020, pp. 1825–1828, https://doi.org/10.1145/3397271.3401320
[10] D. Mujtaba, N. Mahapatra, FairRecruit: a framework for bias-aware resume screening, ACM Transactions on Fairness, Accountability, and Transparency 3 (2) (2025) 1–25, https://doi.org/10.1145/3654674
[11] M. Raghavan, S. Barocas, J. Kleinberg, K. Levy, Mitigating bias in algorithmic hiring: evaluating claims and practices, in: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, Association for Computing Machinery, New York, NY, USA, 2020, pp. 469–481, https://doi.org/10.1145/3351095.3372828
[12] M. Soleimani, A. Intezari, J. Arrowsmith, D.J. Pauleen, N. Taskin, Reducing AI bias in recruitment and selection: an integrative grounded approach, International Journal of Human Resource Management ahead-of-print (2025) 1–36, https://doi.org/10.1080/09585192.2025.2480617
[13] L. Breiman, Random forests, Mach. Learn. 45 (1) (2001) 5–32, https://doi.org/10.1023/A:1010933404324
[14] T. Mikolov, I. Sutskever, K. Chen, G.S. Corrado, J. Dean, Distributed representations of words and phrases and their compositionality, in: C. Burges, L. Bottou, M. Welling, Z. Ghahramani, K. Weinberger (Eds.), Advances in Neural Information Processing Systems, 26, Curran Associates, Inc., 2013, https://proceedings.neurips.cc/paper_files/paper/2013/file/9aa42b31882ec039965f3c4923ce901b-Paper.pdf.
[15] J. Pennington, R. Socher, C.D. Manning, GloVe: global vectors for word representation, in: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2014, pp. 1532–1543, https://doi.org/10.3115/v1/D14-1162
[16] Y. Kim, Convolutional neural networks for sentence classification, in: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2014, pp. 1746–1751, https://doi.org/10.3115/v1/D14-1181
[17] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780, https://doi.org/10.1162/neco.1997.9.8.1735
[18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L.U. Kaiser, I. Polosukhin, Attention is all you need, in: I. Guyon, U.V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), Advances in Neural Information Processing Systems, 30, Curran Associates, Inc., 2017, https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.
[19] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, V. Stoyanov, RoBERTa: a robustly optimized BERT pretraining approach, CoRR arxiv:1907.11692, 2019.
[20] P. He, X. Liu, J. Gao, W. Chen, DeBERTa: decoding-enhanced BERT with disentangled attention, in: International Conference on Learning Representations, 2021, https://openreview.net/forum?id=XPZIaotutsD.

[21] O. Haddad, M.N. Omri, Big textual data analytics using transformer-based deep learning for decision making, in: Advances in Computational Collective Intelligence, Vol. 1234 of Communications in Computer and Information Science, Springer Nature Switzerland, Cham, 2024, pp. 119–131, https://doi.org/10.1007/978-3-031-70234-5_10.

[22] O. Haddad, M.N. Omri, An innovative method for improving sentiment analysis in large language models, Procedia Comput. Sci. 270 (2025) 4605–4614, 29th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2025). https://doi.org/10.1016/j.procs.2025.09.586, https://www.sciencedirect.com/science/article/pii/S1877050925032594.

[23] Y. Tay, M. Dehghani, D. Bahri, D. Metzler, Efficient transformers: a survey, ACM Comput. Surv. 55 (6) (Dec 2022), https://doi.org/10.1145/3530811

[24] S. Minaee, N. Kalchbrenner, E. Cambria, N. Nikzad, M. Chenaghlu, J. Gao, Deep learning–based text classification: a comprehensive review, ACM Comput. Surv. 54 (3) (Apr 2021), https://doi.org/10.1145/3439726

[25] X. Qiu, T. Sun, Y. Xu, Y. Shao, N. Dai, X. Huang, Pre-trained models for natural language processing: a survey, Sci. China Technol. Sci. 63 (10) (2020) 1872–1897, https://doi.org/10.1007/s11431-020-1647-3

[26] A.L. Hunkenschroer, C. Luetge, Ethics of AI-enabled recruiting and selection: a review and research agenda, J. Bus. Ethics 178 (4) (2022) 977–1007, https://doi.org/10.1007/s10551-022-05049-6

[27] H. Chen, Q. Wang, Y. Zhao, GraphResume: enhancing resume classification with skill relational graphs, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, 2025, pp. 2200–2210, https://doi.org/10.18653/v1/2025.emnlp-main.135

[28] Z. Sun, H. Yu, X. Song, R. Liu, Y. Yang, D. Zhou, MobileBERT: a compact task-agnostic BERT for resource-limited devices, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, 2020, pp. 2158–2170, https://doi.org/10.18653/v1/2020.acl-main.195

[29] S. Goyal, A.R. Choudhury, S. Raje, V. Chakaravarthy, Y. Sabharwal, A. Verma, PoWER-BERT: accelerating BERT inference via progressive word-vector elimination, in: Proceedings of the 37th International Conference on Machine Learning, Vol. 119 of Proceedings of Machine Learning Research, PMLR, 2020, pp. 3690–3699, https://proceedings.mlr.press/v119/goyal20a.html.

[30] J. Lee, R. Patel, S. Kim, Explainable AI for human resource decision support, in: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, 2025, pp. 1–15, https://doi.org/10.1145/3544548.3580692

[31] M. Feldman, S.A. Friedler, J. Moeller, C. Scheidegger, S. Venkatasubramanian, Certifying and removing disparate impact, in: Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD ’15, Association for Computing Machinery, New York, NY, USA, 2015, pp. 259–268, https://doi.org/10.1145/2783258.2783311.

[32] M. Hardt, E. Price, N. Srebro, Equality of opportunity in supervised learning, in: D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, R. Garnett (Eds.), Advances in Neural Information Processing Systems, 29, Curran Associates, Inc., 2016, https://proceedings.neurips.cc/paper_files/paper/2016/file/6a9659feb1216f14f7384ba499518b38-Paper.pdf.

[33] D. Pessach, E. Shmueli, A review on fairness in machine learning, ACM Comput. Surv. 55 (3) (Feb 2022), https://doi.org/10.1145/3494672

[34] S. Desai, G. Durrett, Calibration of pre-trained transformers, in: B. Webber, T. Cohn, Y. He, Y. Liu (Eds.), Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, Online, 2020, pp. 295–302, https://doi.org/10.18653/v1/2020.emnlp-main.21.

[35] T. Dao, D. Fu, S. Ermon, A. Rudra, C. Ré, FlashAttention: fast and memory-efficient exact attention with IO-awareness, in: S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, A. Oh (Eds.), Advances in Neural Information Processing Systems, 35, Curran Associates, Inc., 2022, pp. 16344–16359, https://proceedings.neurips.cc/paper_files/paper/2022/file/67d57c32e20fd0a7a302cb81d36e40d5-Paper-Conference.pdf.

[36] C. Rudin, C. Chen, Z. Chen, H. Huang, L. Semenova, C. Zhong, Interpretable machine learning: fundamental principles and 10 grand challenges, Stat. Surv. 16 (none) (2022) 1–85, https://doi.org/10.1214/21-SS133

[37] O. Haddad, M.N. Omri, M.N. Mchiri, Resource-aware handwritten text recognition: a compact CNN-augmentation pipeline for sustainable edge deployment, in: Proceedings of the 18th International Conference on Agents and Artificial Intelligence - Volume 3: ICAART, Insticc, Insticc, 2026, pp. 2471–2478, https://doi.org/10.5220/0014255100004052